



GAMIFIED MOSTON

「&rchitecture Atelier」

Landscape Architecture 2b

CHEN YILI 2026

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RESEARCH QUESTION

How can we apply game design principles to enhance the inclusiveness and interactivity of urban spaces, thereby promoting the social integration of special educational needs student?



RECAP

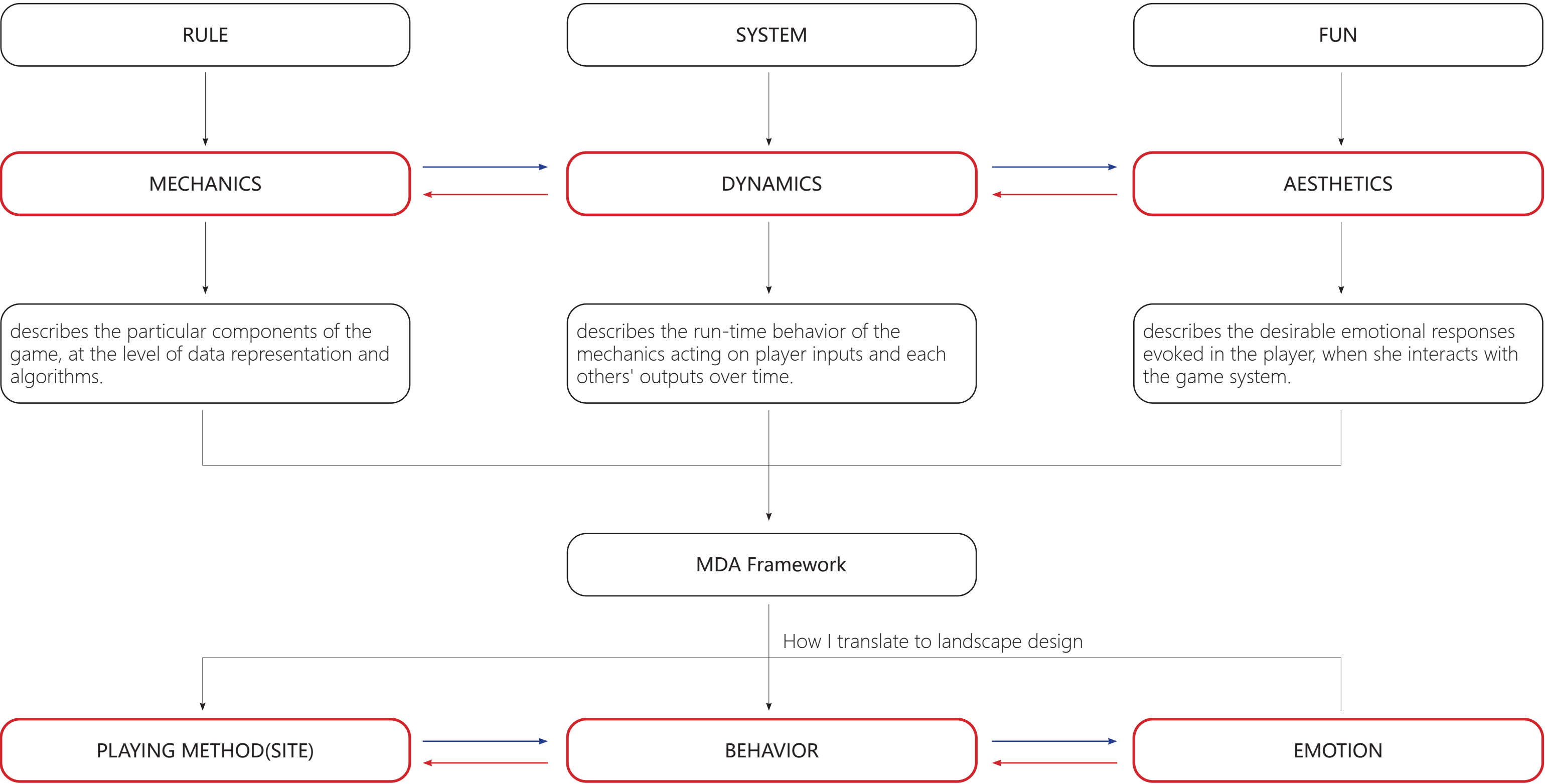
In this section, I will review the content and summary methods I used in the semester 1, so as to better understand the analysis and design I will undertake in semester 2.

PERFORMANCE MAP

	COMMON PROPOSAL 01	COMMON PROPOSAL 02	COMMON PROPOSAL 03	COMMON PROPOSAL 04	COMMON PROPOSAL 05
Performance	 PERFORMANCE 01	 PERFORMANCE 02	 PERFORMANCE 03	 PERFORMANCE 04	 PERFORMANCE 05
Summary	Ethical Ambitions My ethical ambition is to design an inclusive and interactive landscape space, enabling the neglected and marginalized groups in the city to be truly seen and supported.	Creative Differences My focus is on SEN students , particularly those children and teenagers who face challenges in sensory, cognitive and communication aspects. Their needs are often overlooked in the daily urban spaces.	Shared Cities My research has revealed that the streets, school routes and public spaces in Moston can cause sensory overload and movement disorders for SEN students, and these are the key areas where I intervene in the design.	Common Purposes I hope to transform Moston into a supportive and inclusive urban environment that is accessible to SEN students and the wider community by introducing game-like guidance, feedback and immersive interaction.	Just Resources I applied the game design principles based on MDA, attempting to transform the daily urban spaces into playable and experiential urban spaces in order to solve the problems faced by SEN students in the city.

The spatial performance in the first semester focused on the urban landscape of the Moston area in Manchester, UK, exploring how the daily public spaces systematically marginalized the students with special educational needs (SEN) in the local community. Through collage, first-person video recordings, noise maps, heart rate maps and on-site sensory observations, the project transformed these invisible urban barriers into visible spatial evidence. Based on this, the project used the MDA game design framework as the theoretical foundation, proposing Gamified Moston - translating game design principles into landscape strategies, and optimizing the design through game simulation, transforming the daily city into a readable, interactive and emotionally supportive urban space.

MDA: A FORMAL APPROACH TO GAME DESIGN AND GAME RESEARCH *Hunicke, R., LeBlanc, M. and Zubek, R. (2004)*

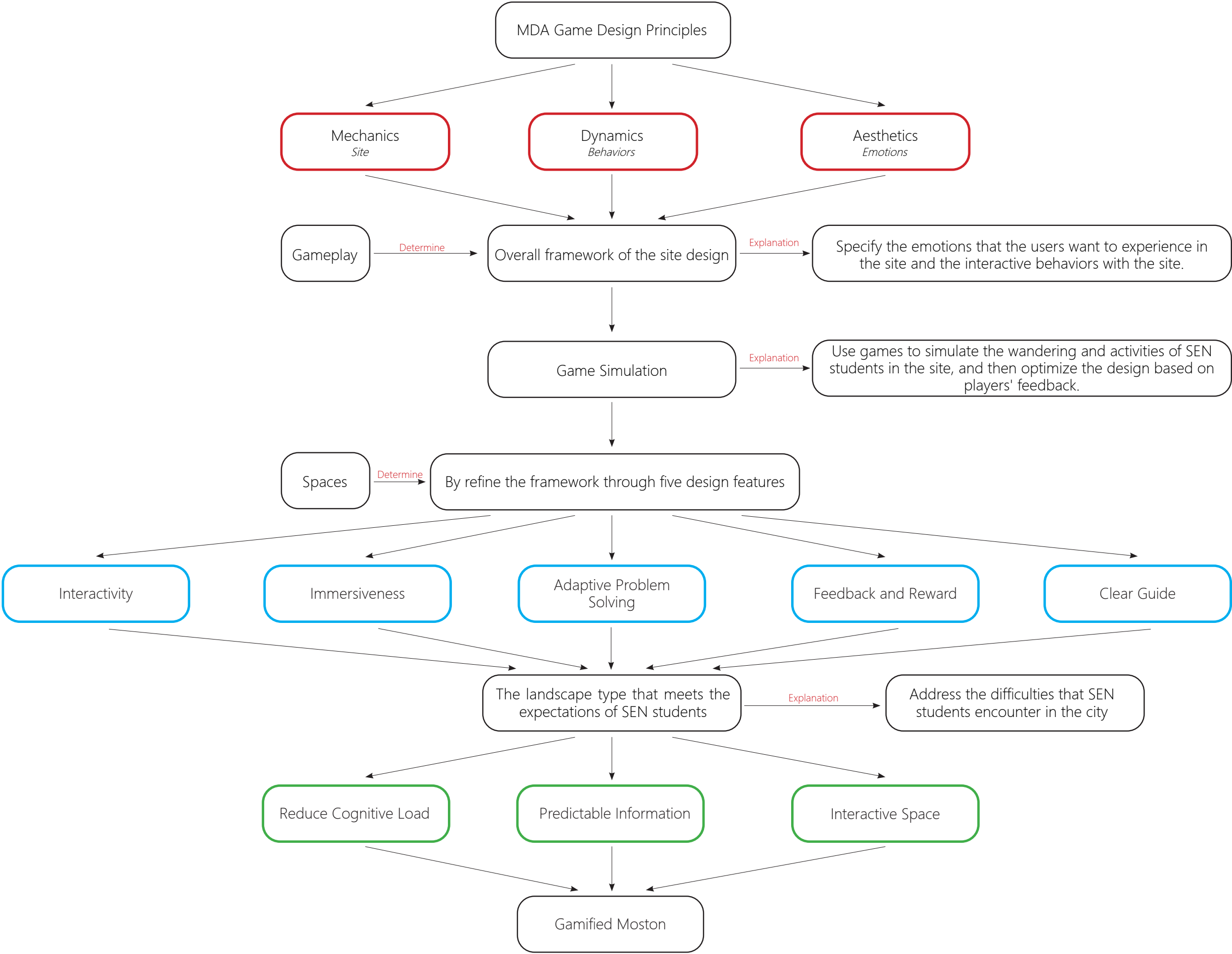


LEGENDS

- Designer's Perspective
- Uer's Perspective

REFLECTION: In game design, the MDA design principles are used to summarize the design of the game. I would like to attempt to translate the mechanisms, dynamics and aesthetics in MDA into elements in the landscape to assist in the design. The mechanisms correspond to the sites and gameplay in the landscape, while the sites support human activities. And dynamics correspond to behaviors. Behaviors determine the activities of people in the space. Aesthetics corresponds to emotions. Different behaviors between people or between people and the environment give rise to various emotions.

DESIGN METHOD





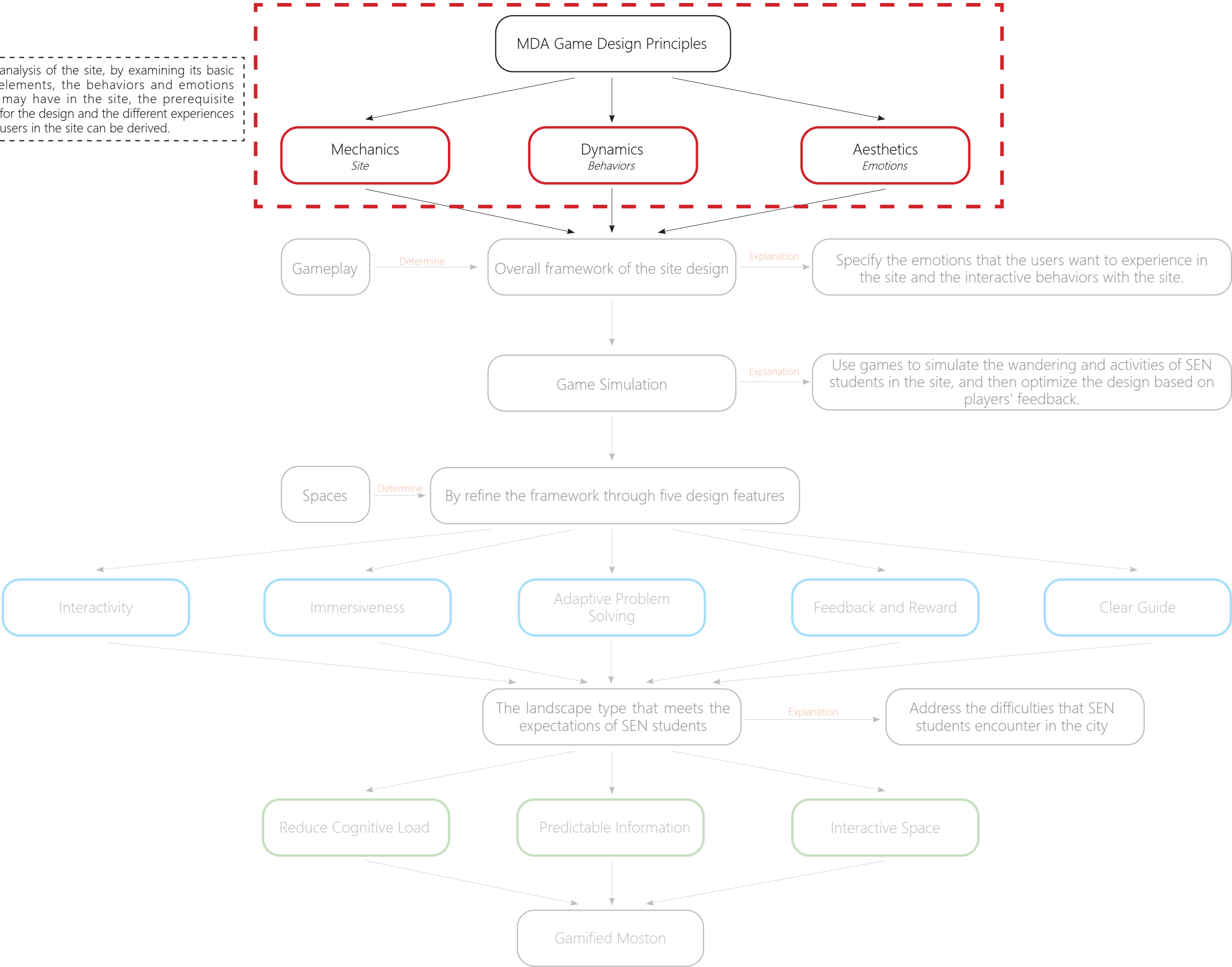
AESTHETICS

EMOTIONS

Aesthetics corresponds to emotions in a landscape design: as users generate different dynamics (behaviors) under spatial guidance, they experience different experiential outcomes—i.e., emotions. Design shapes these dynamics through spatial mechanisms, and users' behaviors and perceptions then work together to produce the emotional outcomes that can be felt in the landscape.

DESIGN METHOD

Through the analysis of the site, by examining its basic constituent elements, the behaviors and emotions that people may have in the site, the prerequisite requirements for the design and the different experiences of the design users in the site can be derived.

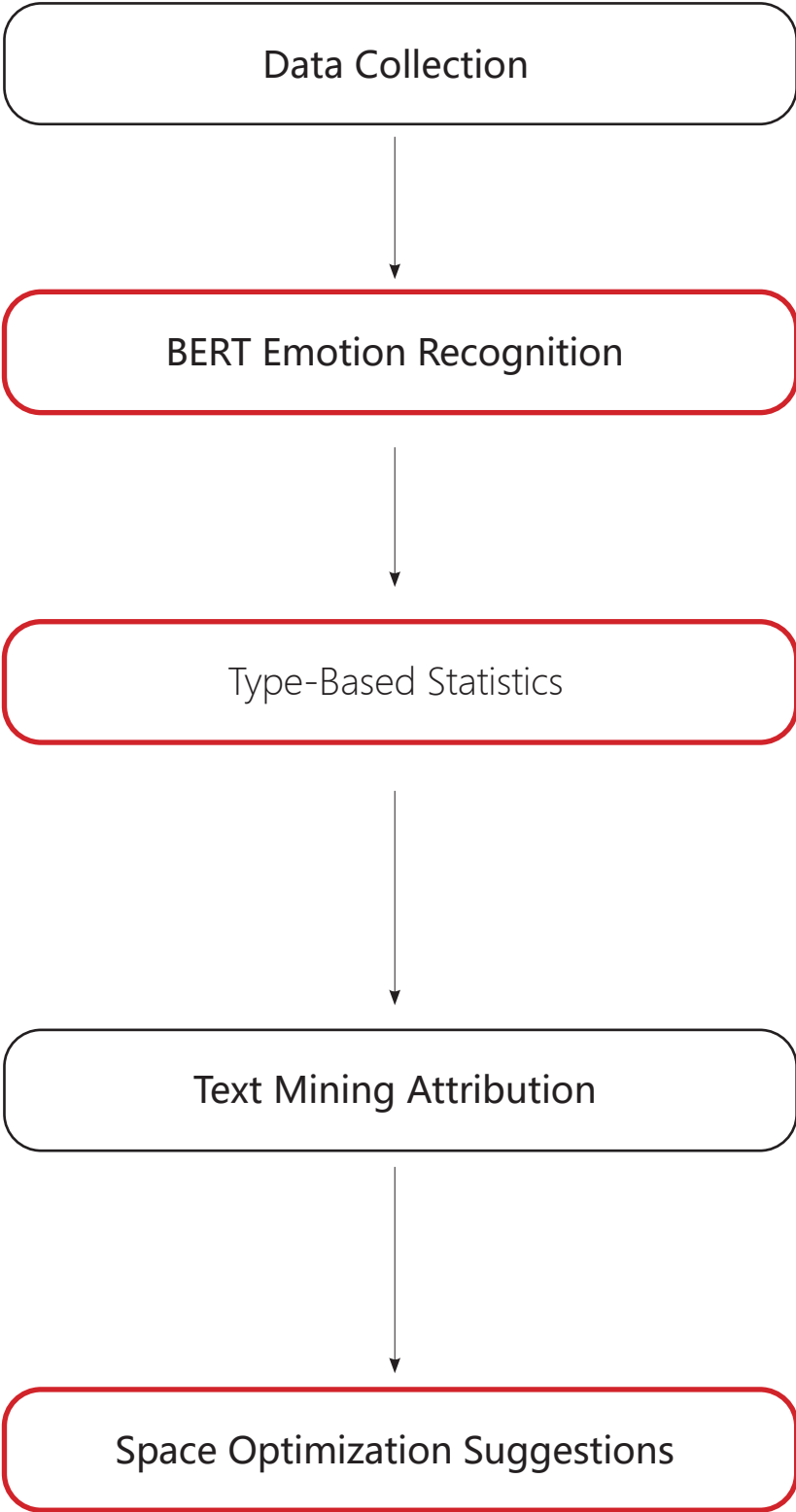


EVALUATING EMOTIONAL PERCEPTION OF SPATIAL HOTSPOTS VIA DEEP LEARNING: A CASE STUDY OF SHANGHAI *Lu-ming, C. U. I., Ling-yan, Q. U., & Dan, H. (2021)*

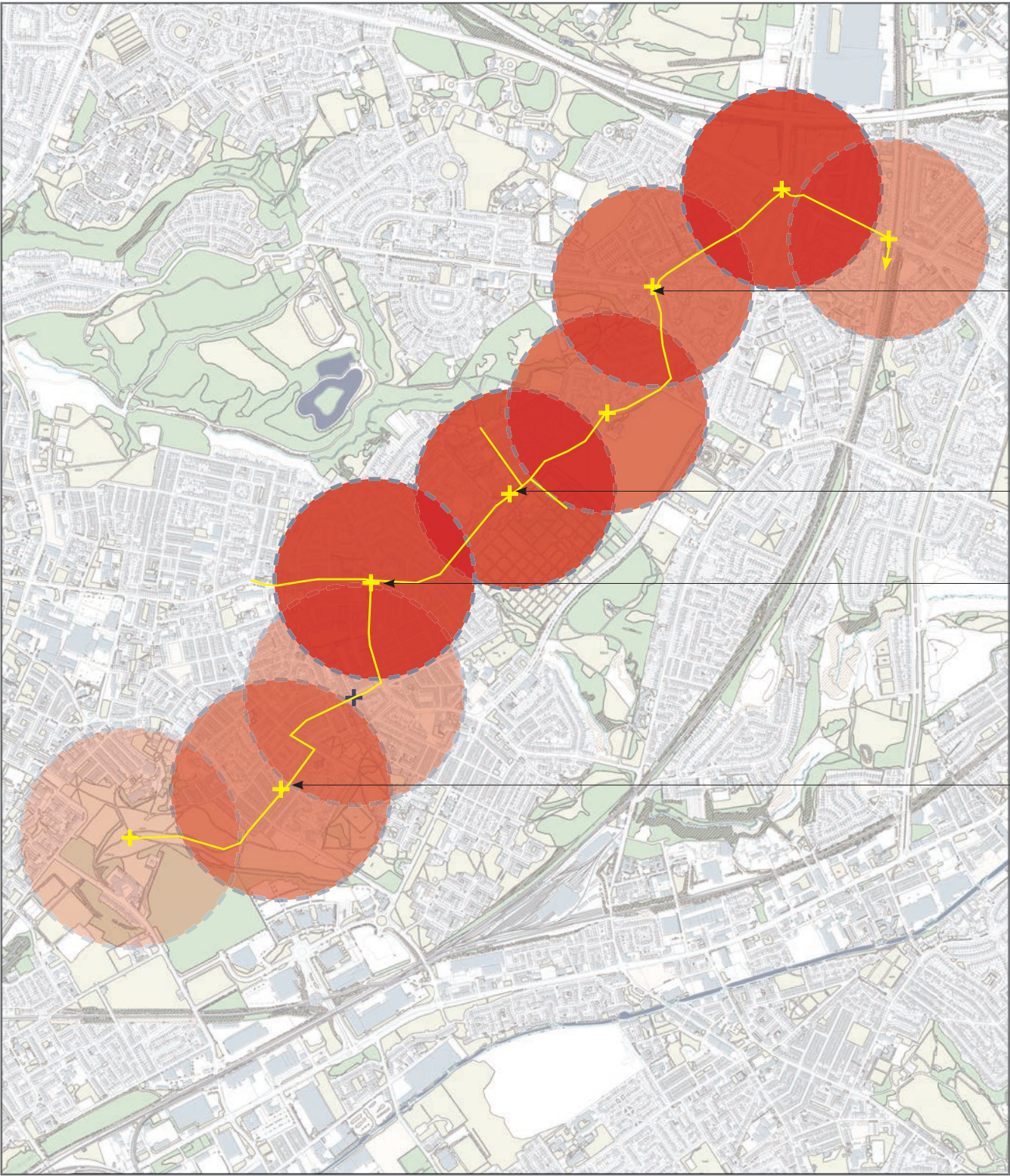
This research has established a data-based emotion perception method, which integrates data collection, Python data preprocessing, multi-class emotion classification deep learning based on BERT, GIS spatial mapping (point of interest distribution, kernel density, emotion map), and text mining (comment extraction + lexical analysis) to identify the spatial and emotional relationships and key physical environmental factors.

In the subsequent analysis, I will apply this method: Firstly, collect on-site sensory data and user behavior observations; then map the emotions onto the site planning diagram to visually identify the hotspots of negative perception; next, use spatial overlay analysis to link these emotions with physical mechanisms; Finally, utilize the identified causal relationships to optimize the design intervention measures.

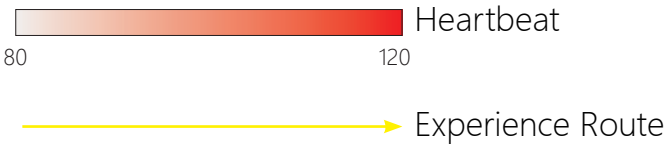
Lu-ming, C. U. I., Ling-yan, Q. U., & Dan, H. (2021)



THE HEARTBEAT MAP OF MOSTON *Yili Chen. (2026)*



LEGENDS



Physical disability: The sidewalks are always so narrow, and during breaks the crowds are so dense that it's very difficult for me to move forward.

Blind person: Vehicles are parked randomly on the pedestrian path, and the advertisements are blocking my way forward. I start to doubt whether I can walk ahead.

Blind person: The noisy intersection has no traffic lights. The honking sounds from the horns made me hesitant to cross the road.

Physical disability: The curb is too high, preventing me from walking on the sidewalk.



NERVOUS

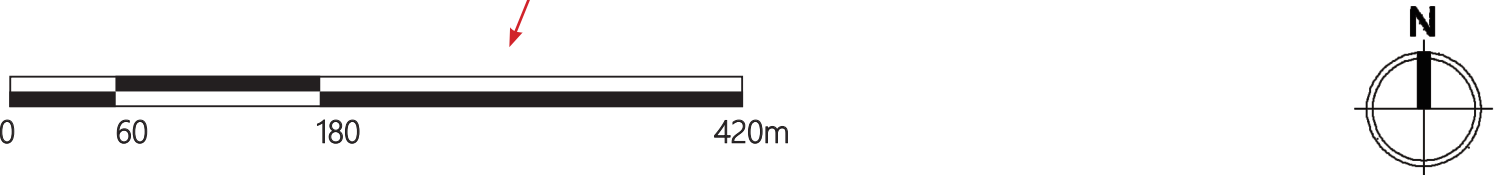
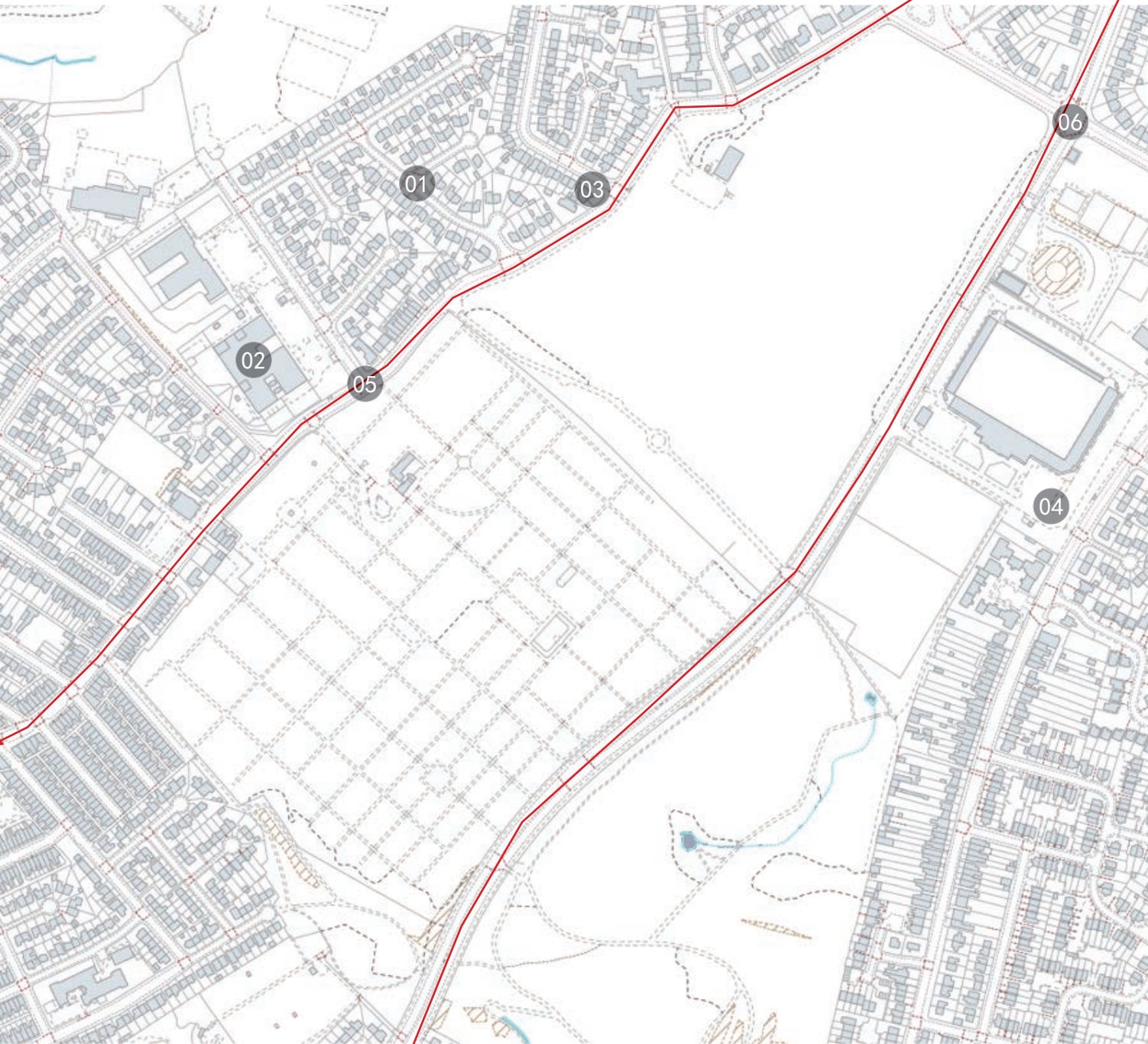
IRRITABLE

AFRAID

CONFUSE

REFLECTION: In Moston, many facilities are not accessible to people with disabilities. The crowded sidewalks are not suitable for those with physical disabilities to walk on. The noisy traffic makes it impossible for the blind to distinguish the road. The sidewalks occupied by vehicles and billboards constantly affect their judgment of whether the road is safe. And my inner state changed gradually from the comfort of walking in the park to the tension and anxiety in the urban space.

MOSTON MAP



REFLECTION: By experiencing the urban space, we can determine the emotions that SEN students will experience in Moston. By analyzing the emotions that SEN students experience in the space, we can transform the previously difficult-to-describe subjective experiences into observable dimensions that can be discussed and compared, thereby conducting a further analysis of the deficiencies in the urban space.

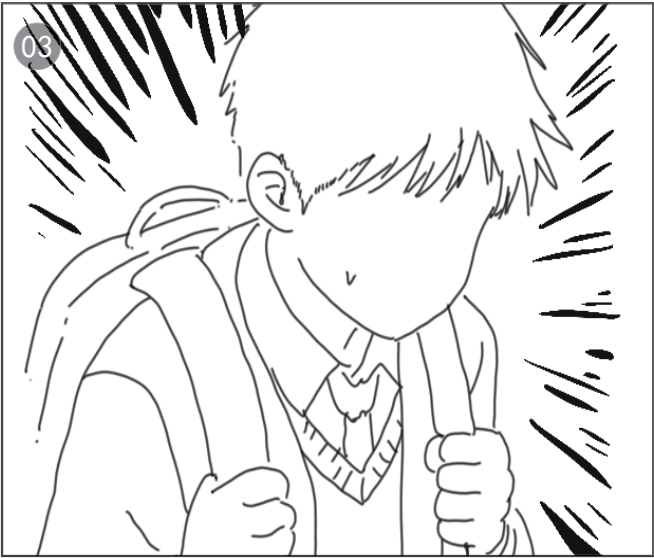
AESTHETICS(EMOTIONS)



Collapse / Fatigue



Frustrated



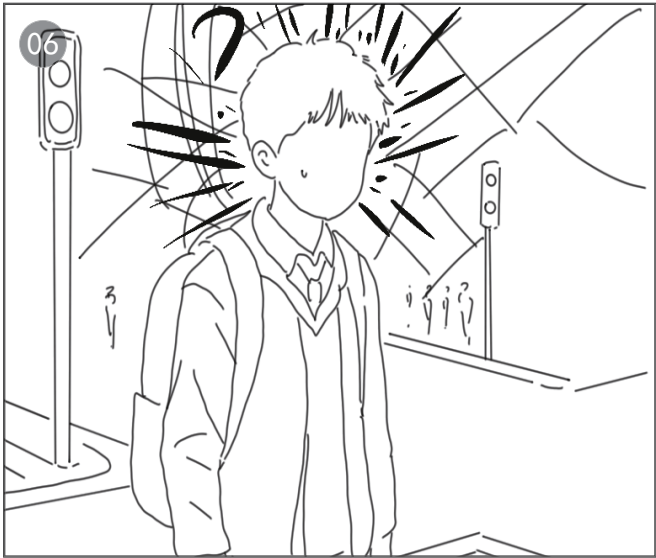
Fear



Nervous



Overload



Confused

NEXT QUESTION

What kind of Dynamics(Behaviors) can lead to such Aesthetics(Emotions)? And how should we improve these emotions?

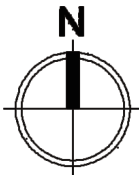
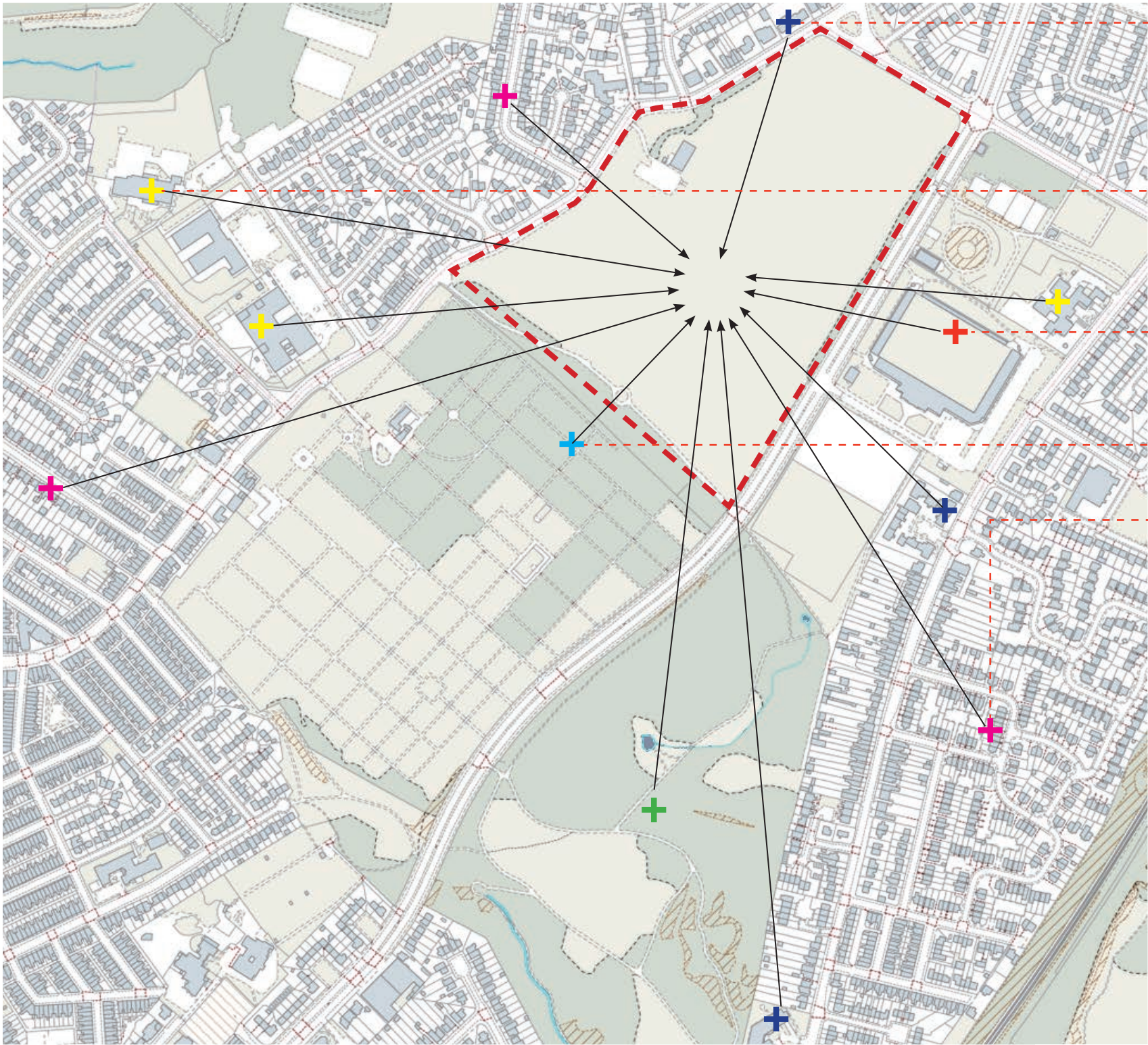


DYNAMICS

BEHAVIORS

Dynamics corresponds to behaviours in landscape design: guided by the Mechanics (site/space), users develop diverse patterns of use within the space, such as movement flows, stopping, and gathering. Designers shape these dynamics by designing the site's spatial mechanisms—such as path organisation, edges and thresholds, activity nodes and amenities, and wayfinding information—thereby influencing people's perceived experience quality and emotional responses to the landscape.

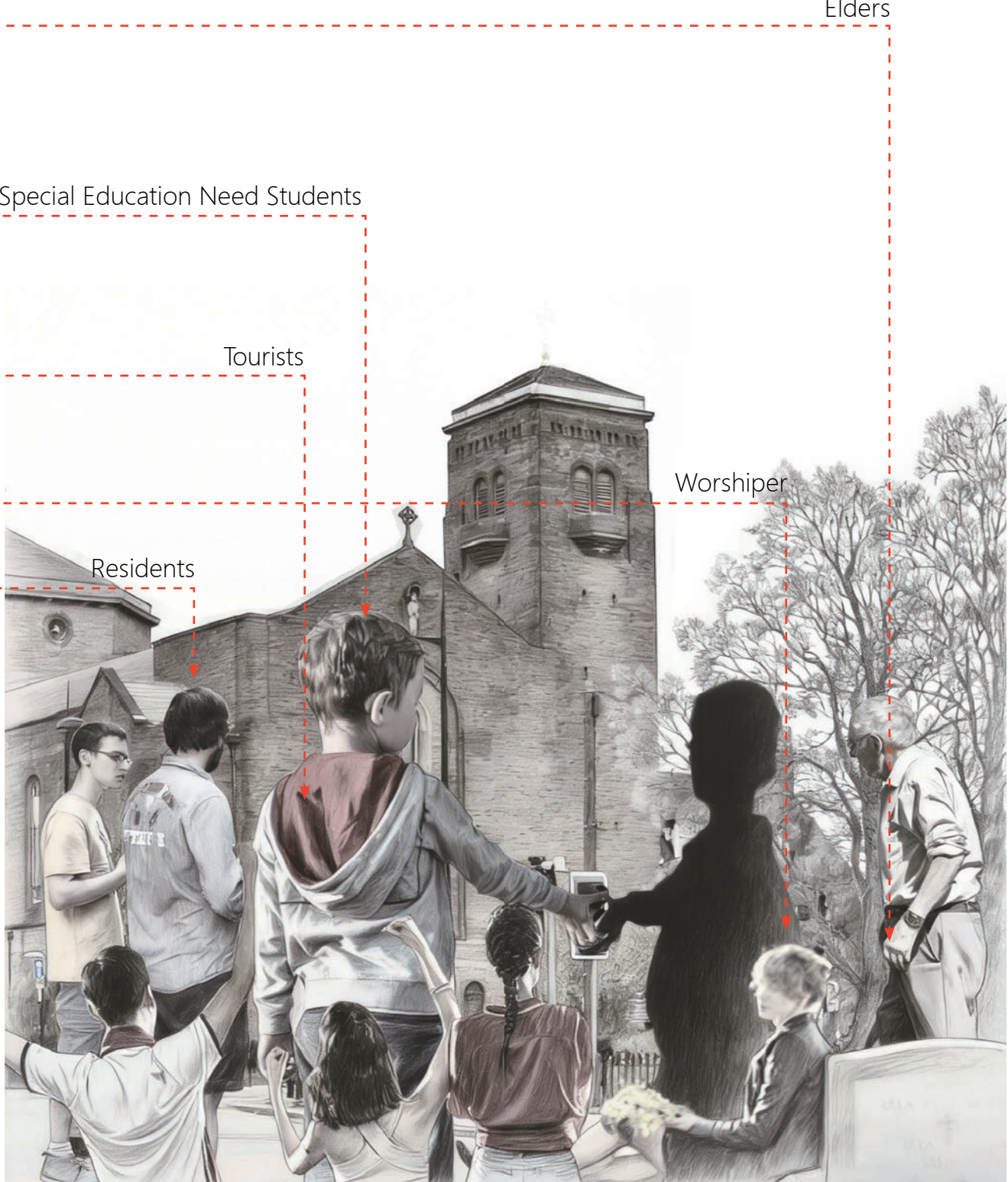
STAKEHOLDER PLAN



LEGENDS

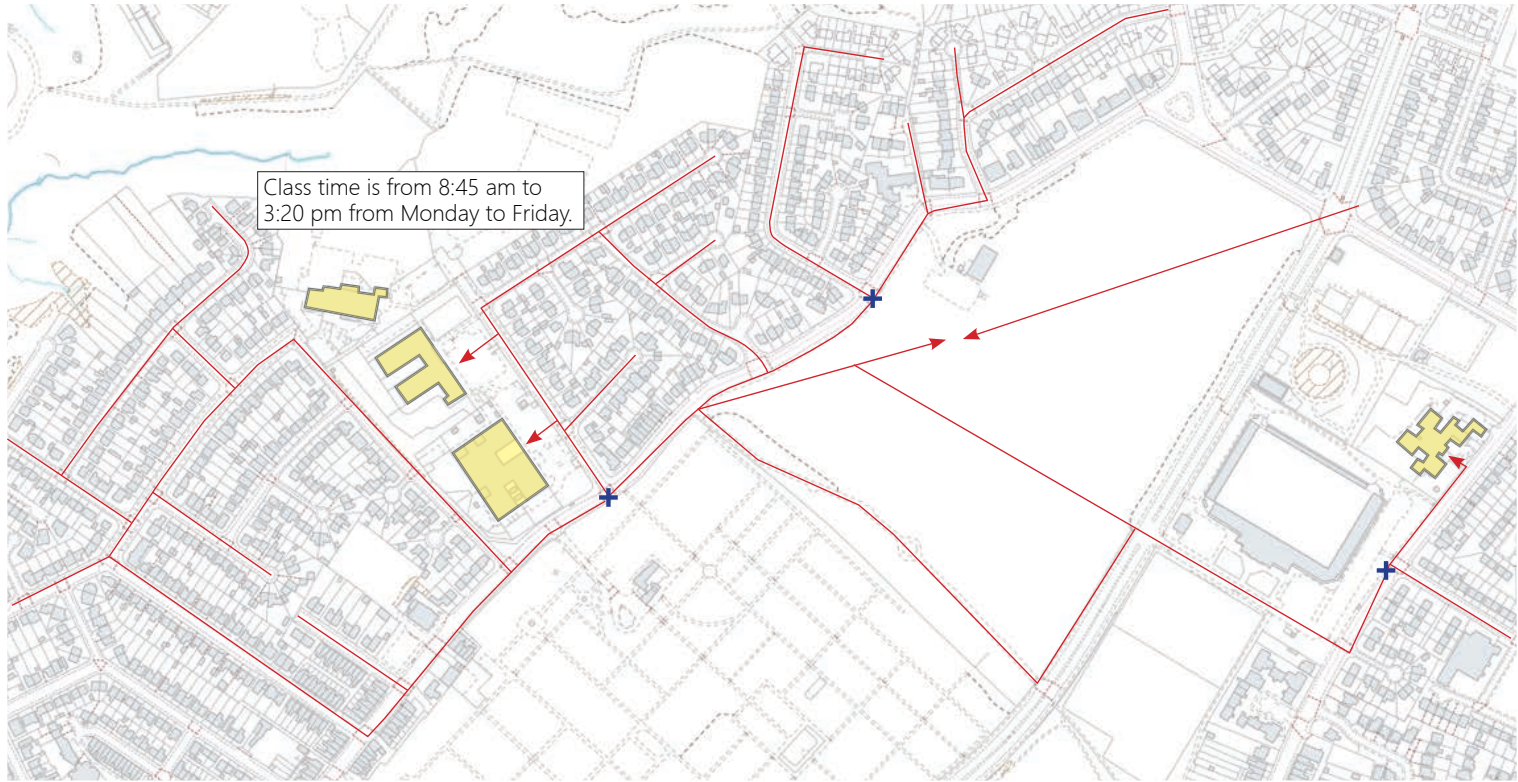
- + School
- + Cemetery
- + Stadium
- + Park
- + Residential Area
- + Nursing Home

STAKEHOLDER COLLAGE

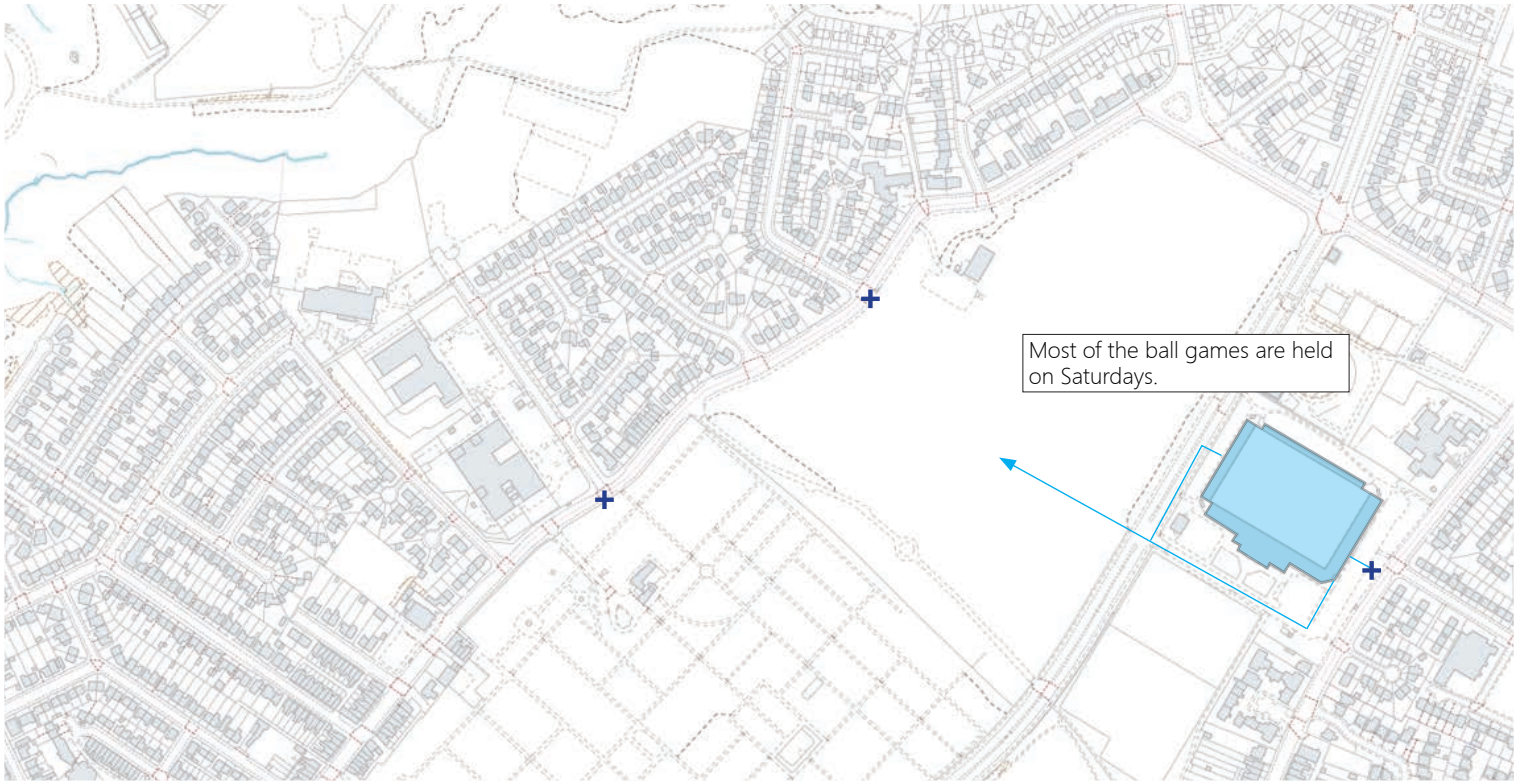


REFLECTION: The surrounding area of the site is inhabited by many different groups of users. As the center of the surrounding plots, the site naturally has the advantage of attracting various groups. In future designs, it is hoped that by catering to the needs of different groups around the site, different groups can be attracted to the site for exchanges, thereby strengthening the mutual communication among different groups in Moston.

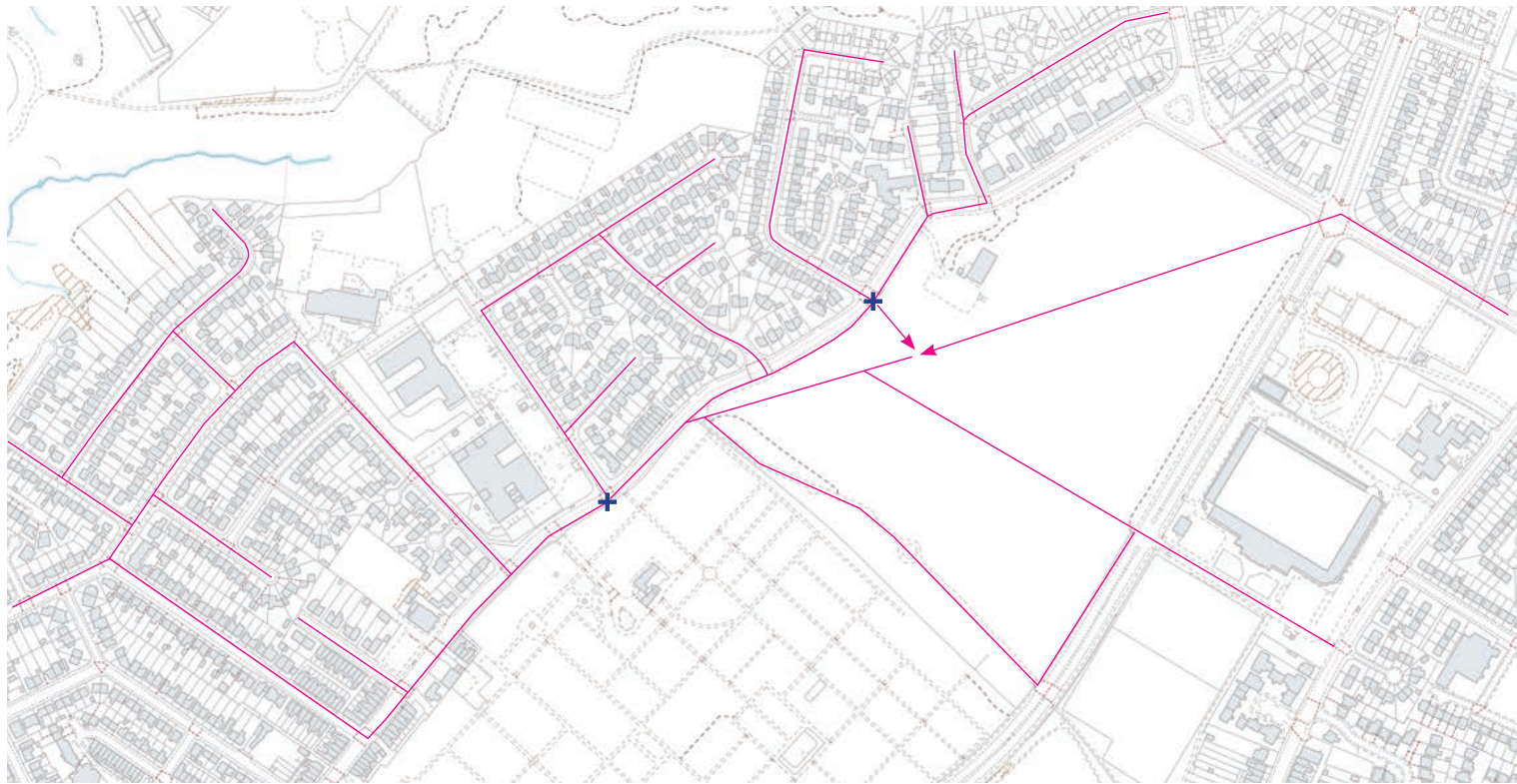
SEN STUDENTS ROUTE



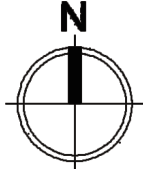
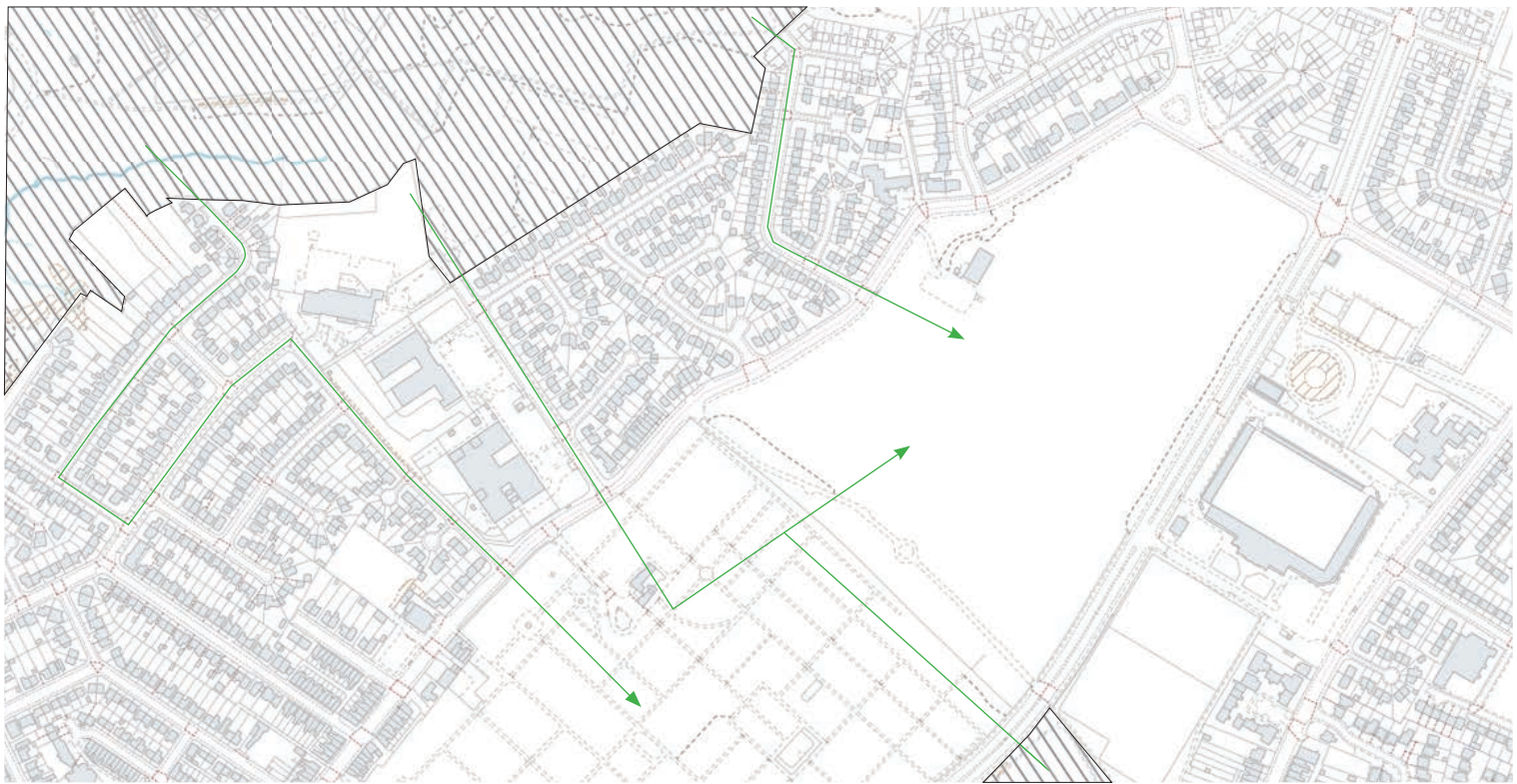
VISITORS ROUTE



RESIDENT ROUTE



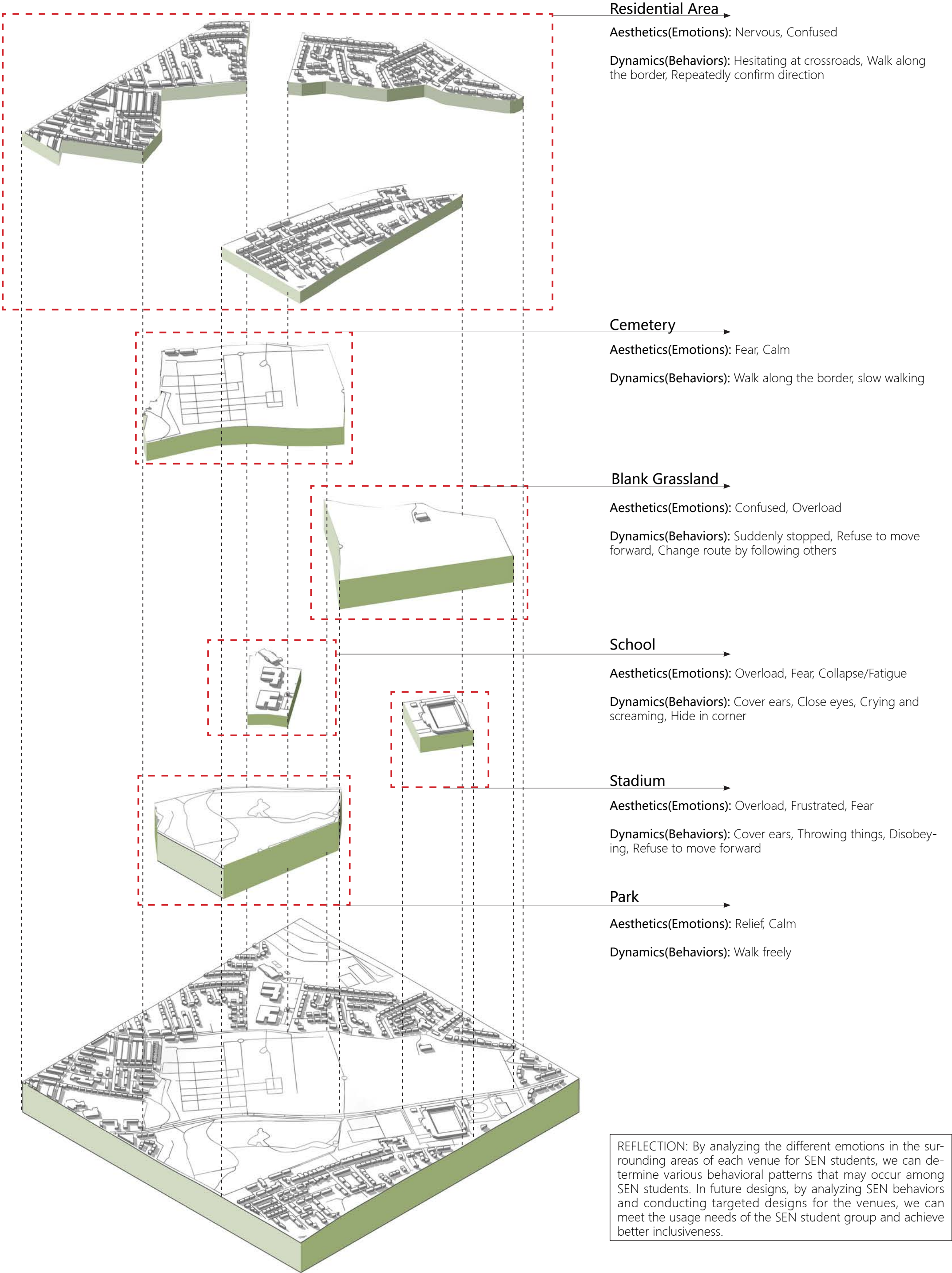
SPECIES ROUTE

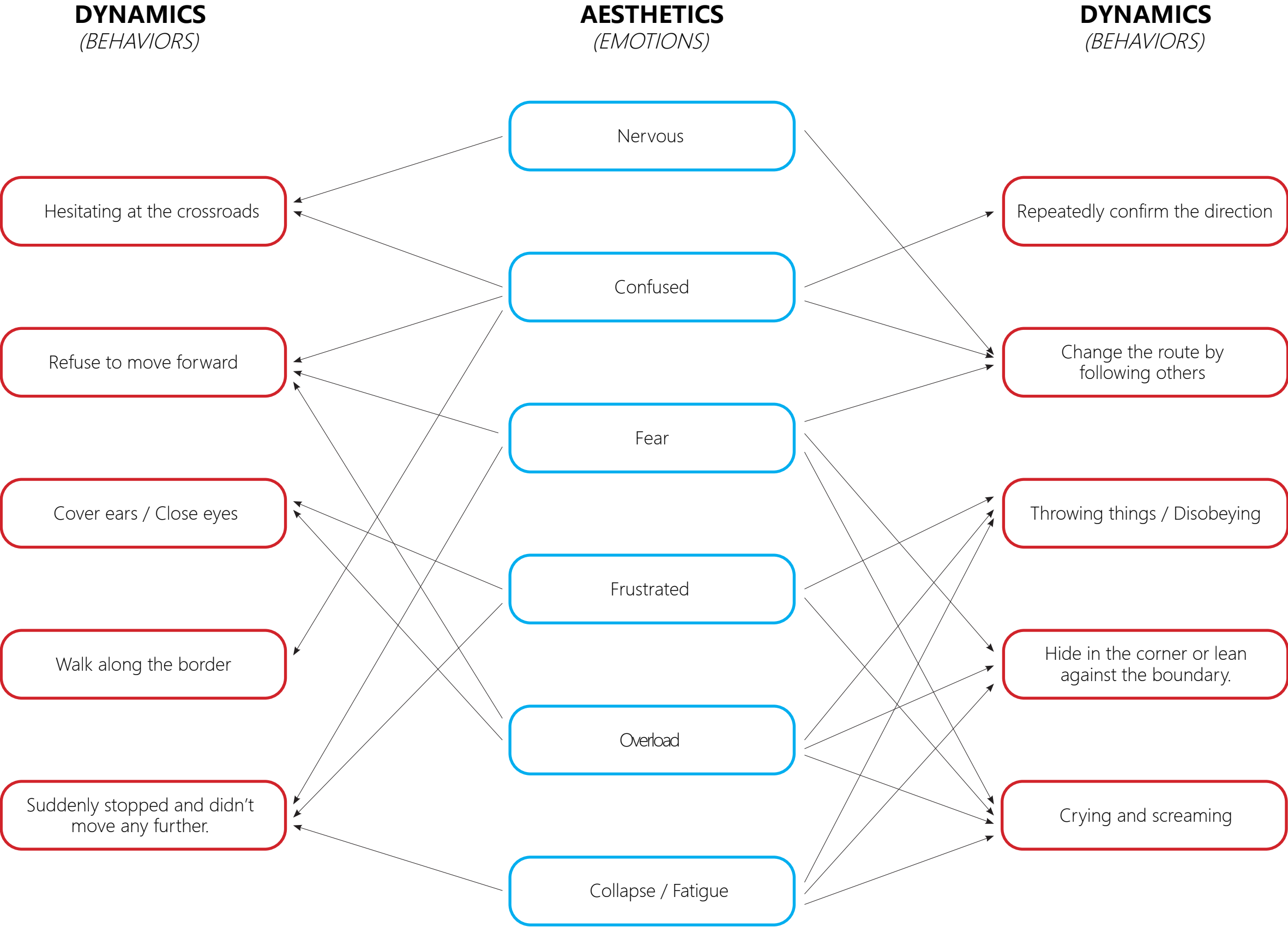


- LEGENDS**
- + School
 - + Cemetary
 - + Stadium
 - + Park
 - + Residential Area
 - + Nursing Home

REFLECTION: The surrounding area of the site is inhabited by many different groups of users. As the center of the surrounding plots, the site naturally has the advantage of attracting various groups. In future designs, it is hoped that by catering to the needs of different groups around the site, different groups can be attracted to the site for exchanges, thereby strengthening the mutual communication among different groups in Moston.

SEN STUDENT BEHAVIORS

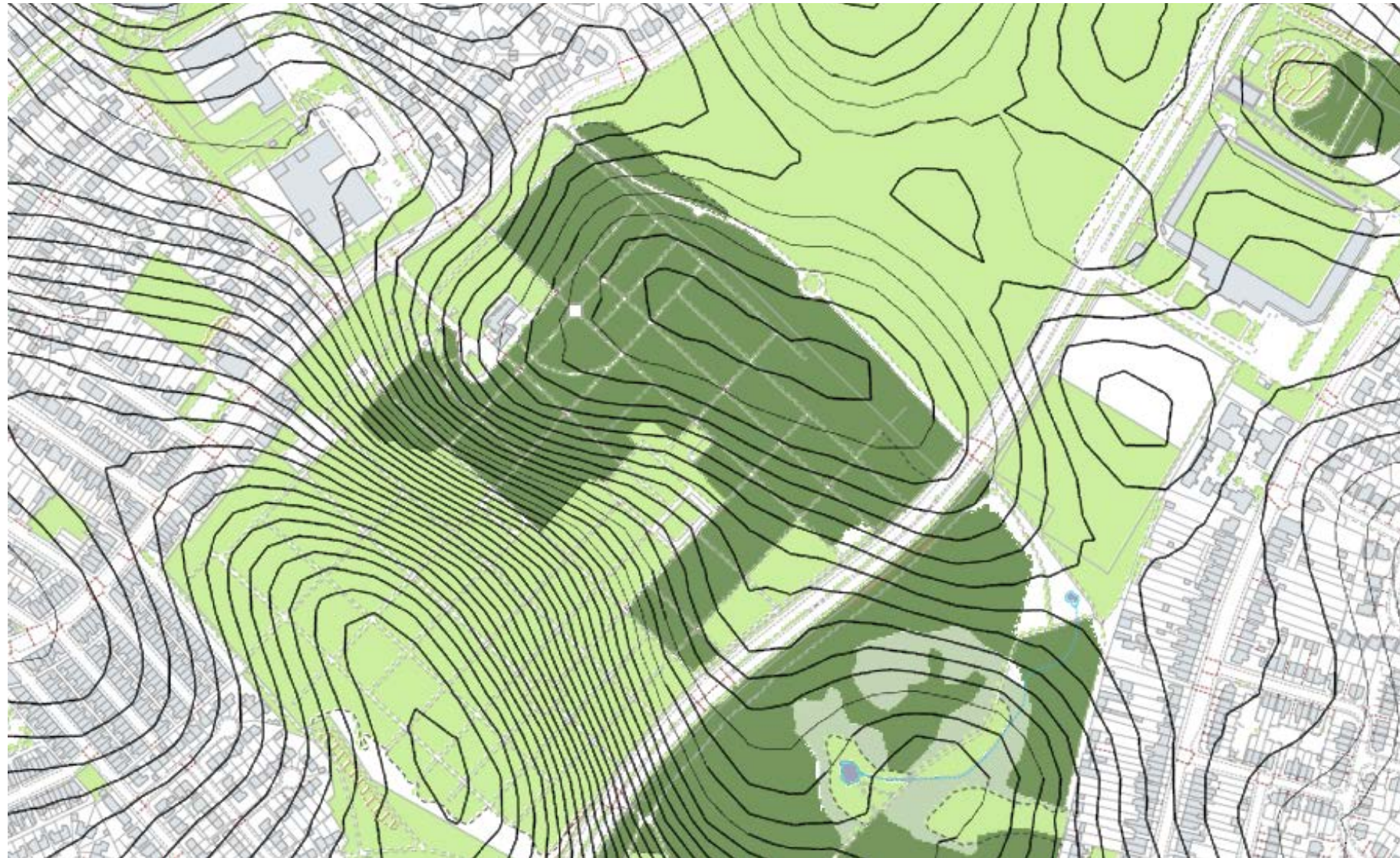




REFLECTION: Starting from Aesthetic (emotion), I analyzed the Dynamic (behaviors) that would occur in the site. By correlating emotions with behaviors, the same emotion can manifest as multiple behaviors, and the same behavior can also correspond to different emotions. Through the transition from emotion to behavior, we can better understand the experiences of SEN students in the site, and thereby conduct the derivation of the design.

NEXT QUESTION

What kind of Mechanics (Sites) determines the emergence of Dynamics (Behaviors) ? How should we change the behaviors and emotions of SEN students in the Moston environment by improving the mechanism?

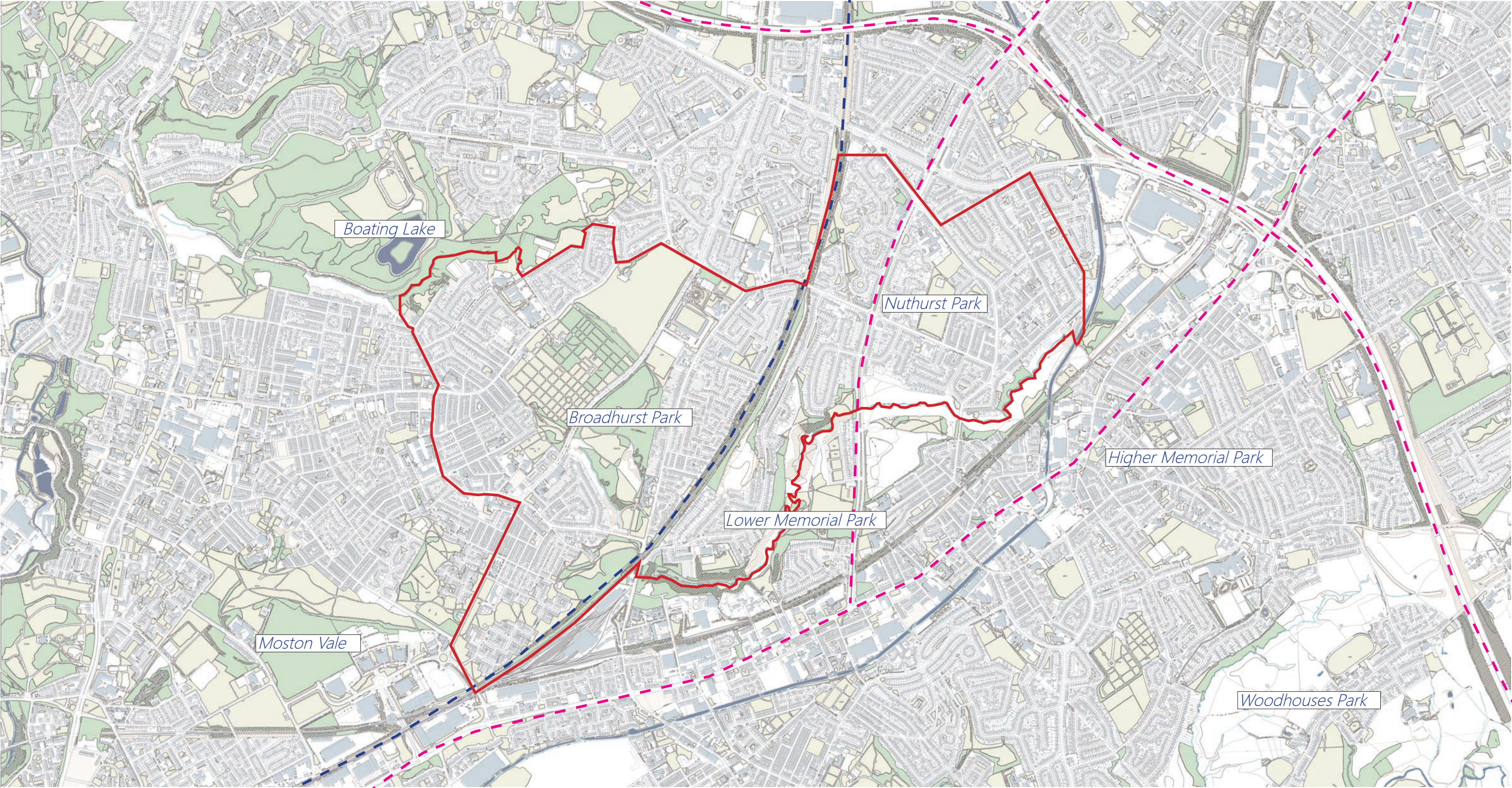


MECHANICS

SITES

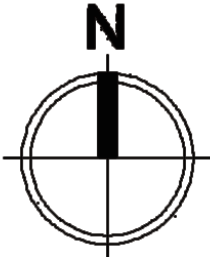
I correlate the "mechanism" in the landscape with the site itself: by analyzing the nature of the site (such as spatial composition, ecological conditions and accessibility, etc.), identifying its key spatial elements and constraints, and thereby deriving and guiding the behavioral patterns and activity modes of specific user groups within that space.

MOSTON SURROUNDING



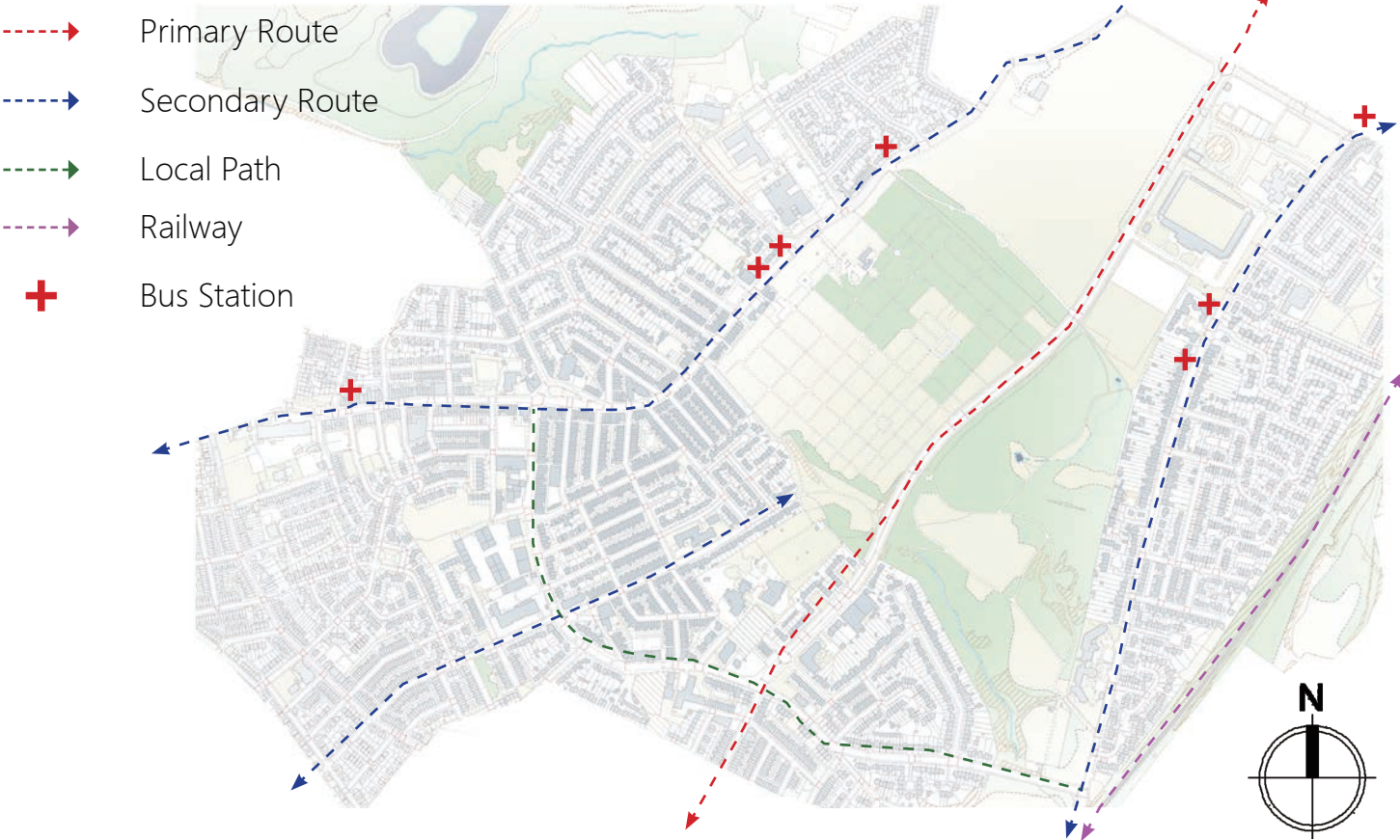
LEGENDS

- Woodland Area
- Grassland Area
- Water Area
- Moston Boundary
- Primary Route
- Railway



REFLECTION: We can observe that the green space resources in the Mosotn area are abundant but lack planning. Large areas of green land are left or used as cemeteries, and people lack playable and active green space areas. Secondly, the transportation in Mosotn is relatively developed and convenient, and it has good development potential.

TRANSPORT



The transportation in Moston is relatively convenient, but the well-developed transportation space not only generates a lot of noise but also divides the green spaces of the city.

FUNCTIONAL



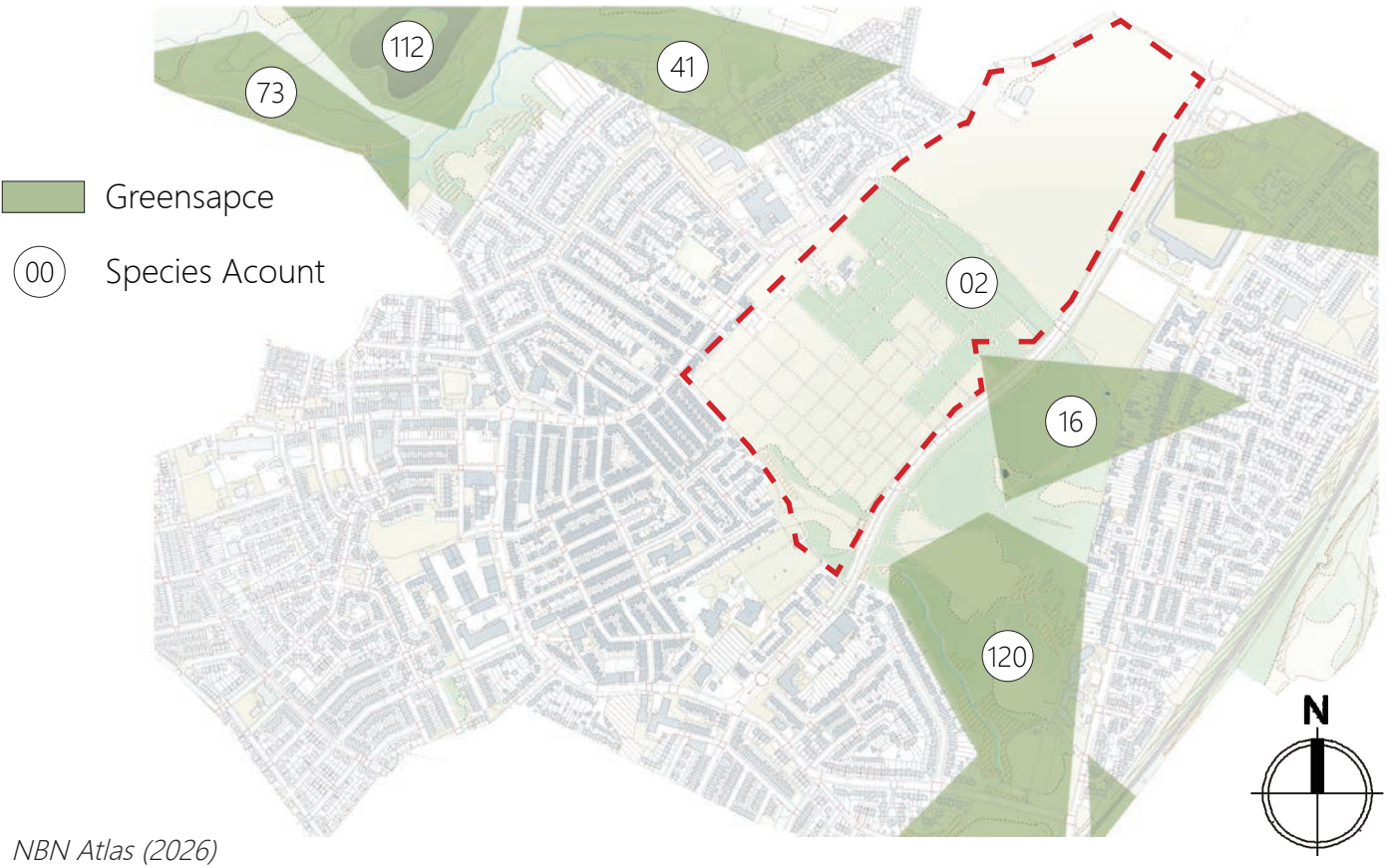
The road passes from the noisy commercial area to the peaceful Grassland, allowing one to have an experience of moving from a bustling urban area to a serene natural region.

LANDUSE



The green spaces in the city are rather fragmented, and most of them are vacant grasslands.

BIODIVERSITY

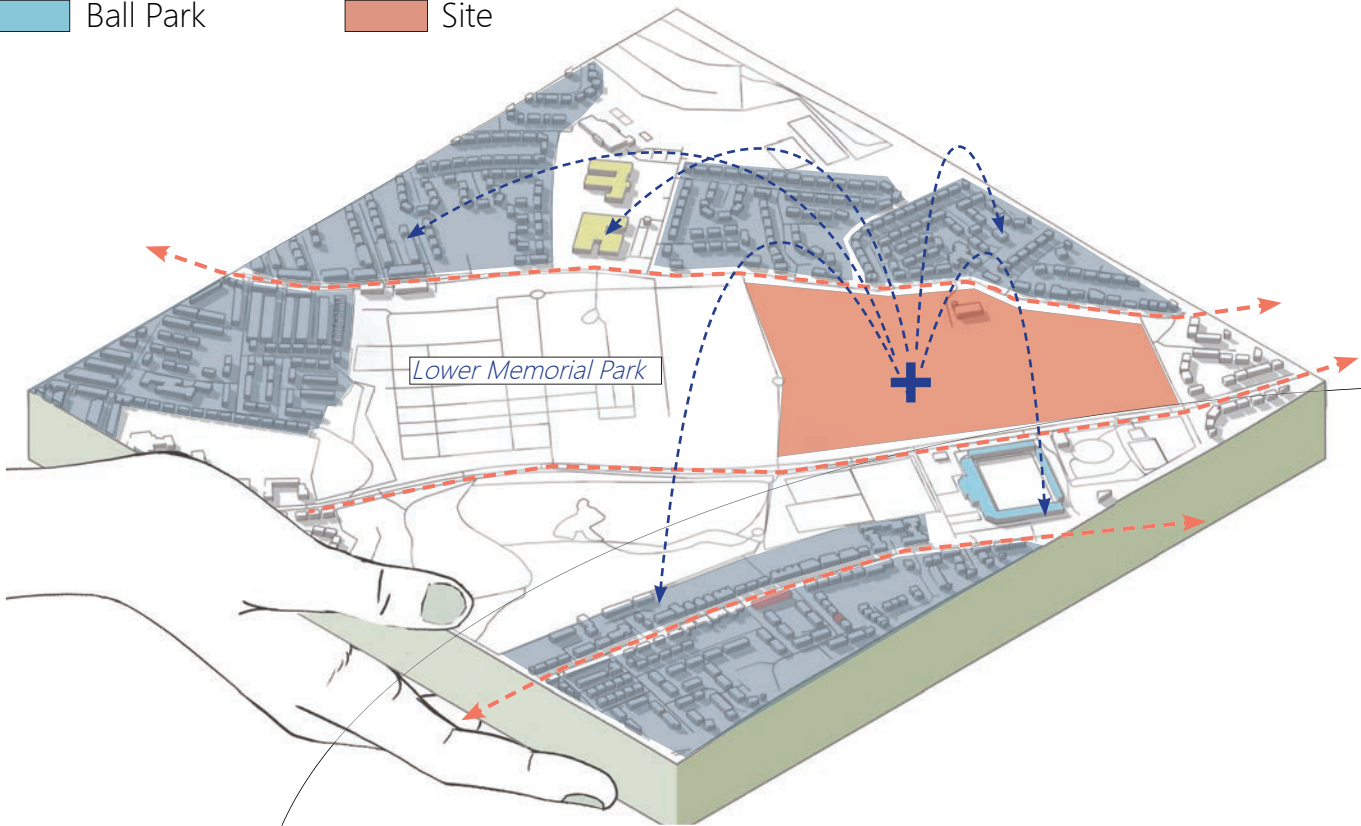


NBN Atlas (2026)

As the green space occupying a large portion of the central area of the city, it has not maintained a high level of biodiversity.

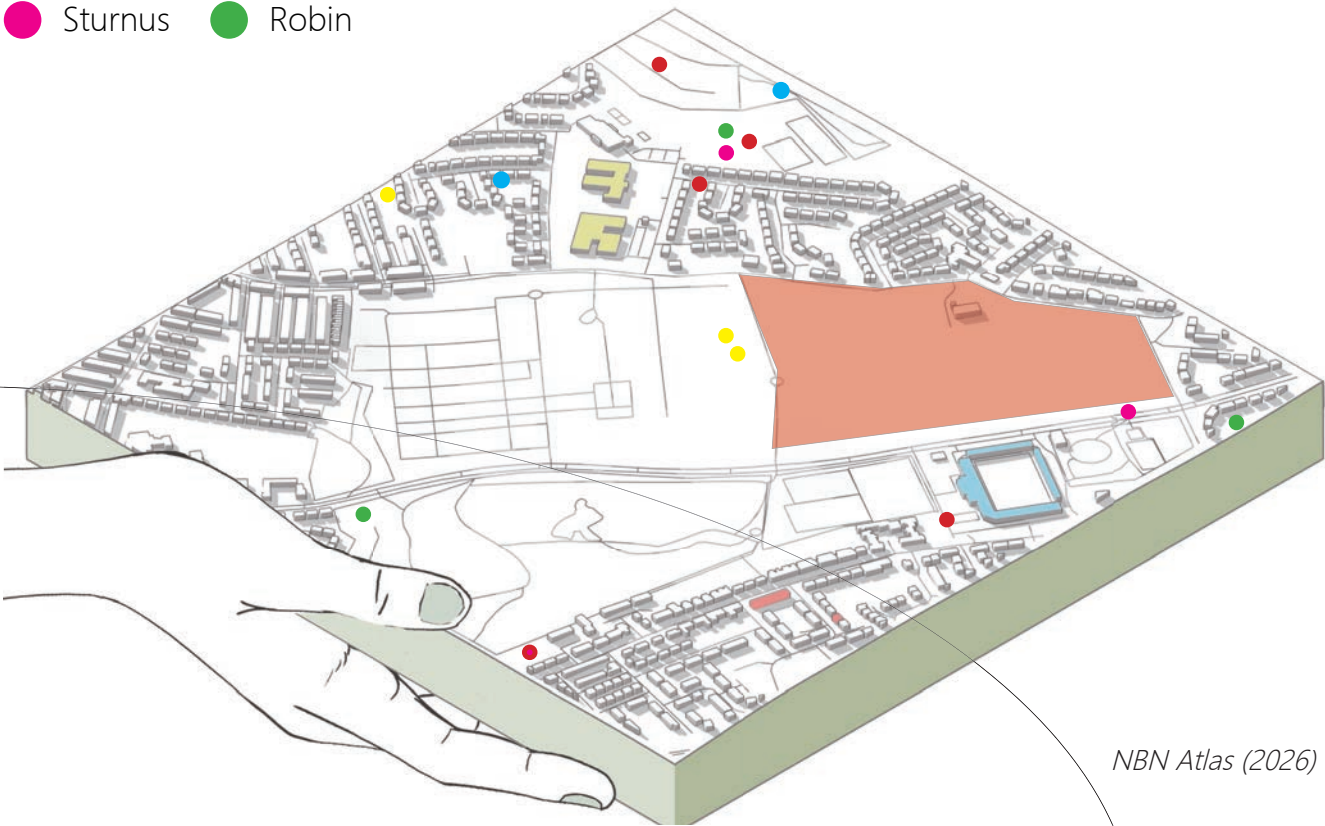
- Residence Area
- School
- Ball Park
- Site

- Picapica
- Hedgehog
- Woodpigeon
- Sturnus
- Robin



FUNCTIONAL

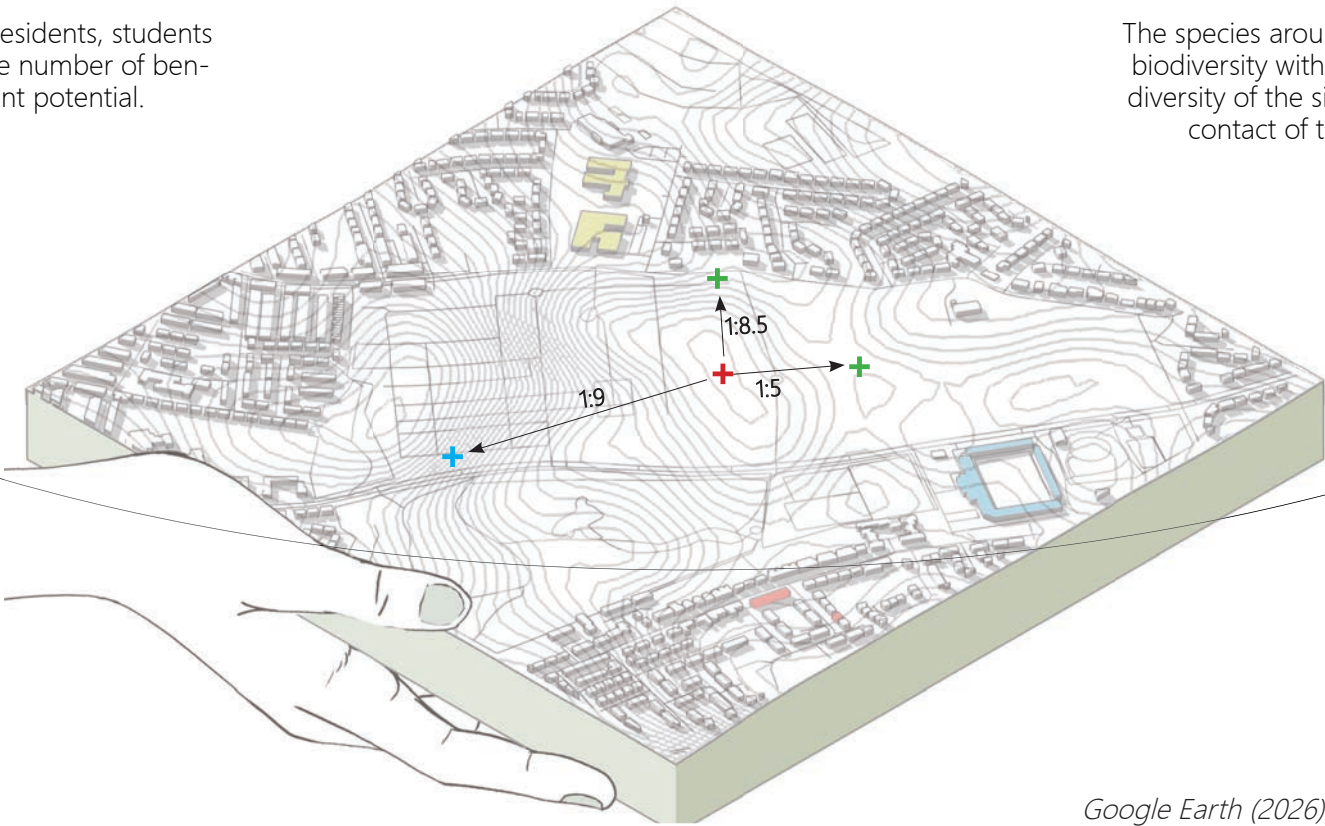
The user groups around the venue include residents, students and visitors to the stadium. There are a large number of beneficiaries and considerable development potential.



NBN Atlas (2026)

BIOLOGICAL MAP

The species around the site are diverse, but there is a lack of biodiversity within the site. Consider enhancing the species diversity of the site while also increasing the interaction and contact of the user group with natural organisms.



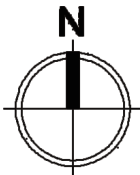
Google Earth (2026)

CONTOUR

The elevation variations within the site are significant. In the later design phase, it is necessary to take into account the elevation differences within the site.

- Highest Point
- Hight Point
- Lowest Point

ECOLOGICAL APPRAISAL_GREENSPACE

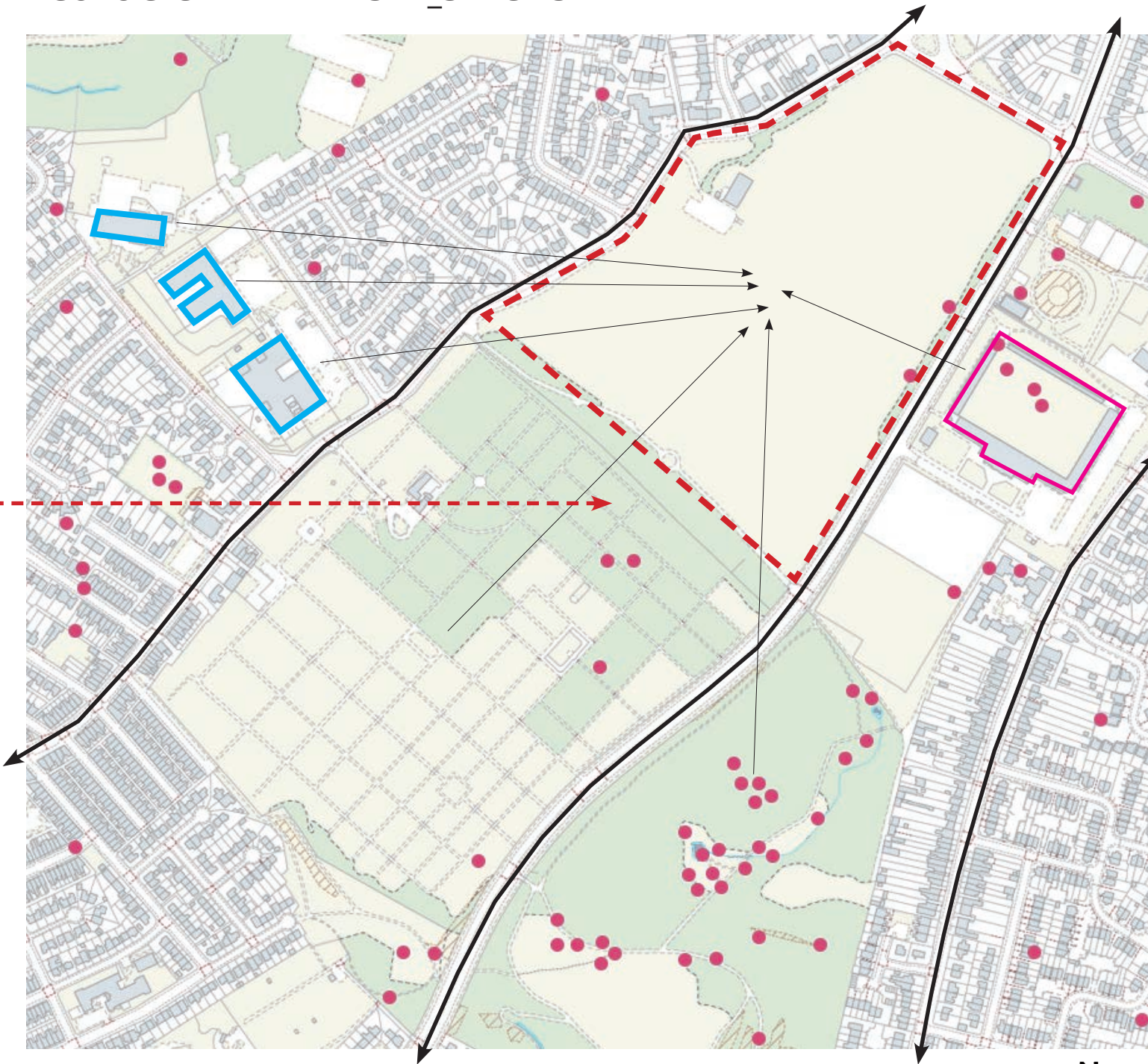


LEGENDS

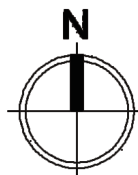
Woodland Wetland Grassland + Highest Point + Hight Point

REFLECTION: By overlaying the site hight deference map and the ecological analysis map, we can Pnd that the highest point of the site is located between the graveyard and the designed site, which may serve as the best viewing spot for the site in the future. At the same time, we can Pnd that the trafPc roads and residential areas have interrupted the connection between ecological green spaces, thus forming fragmentation. In the future design, we will consider repairing fragmentation to increase the ecological diversity of the site.

ECOLOGICAL APPRAISAL_SPECIES



NBN Atlas (2026)



LEGENDS

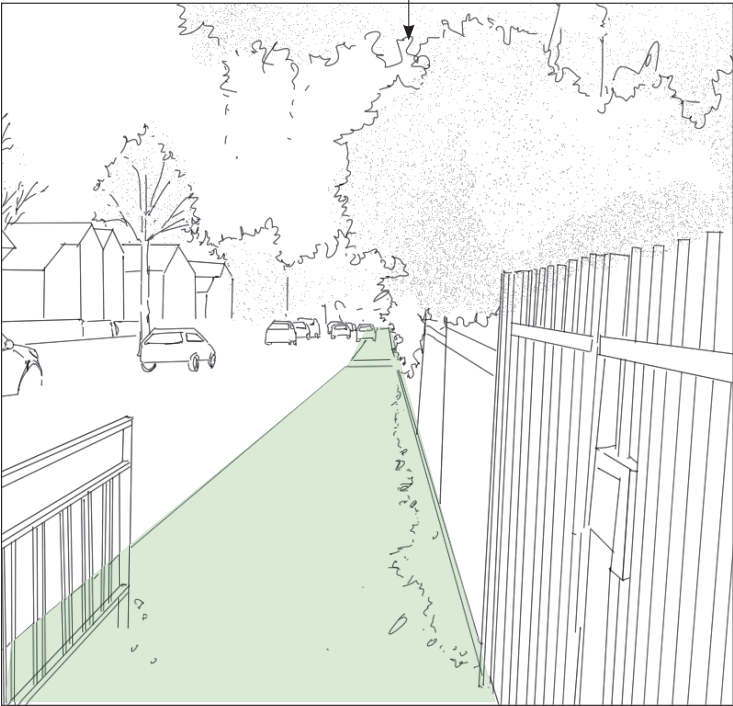
Species

REFLECTION: Through the analysis of species around the site, we can Pnd that the number of species on the site is sparse, but the surrounding green spaces maintain good species diversity. By overlaying the road analysis of Moston, we can Pnd that the trafPc in Moston has divided the connections between green spaces. Secondly, we can Pnd that users on the east side of the site may encounter some difPculties when entering the site, and in future designs, we will also consider strengthening the connection between the east side users and the site.

MOSTON SURROUNDING *Yili Chen. (2026)*

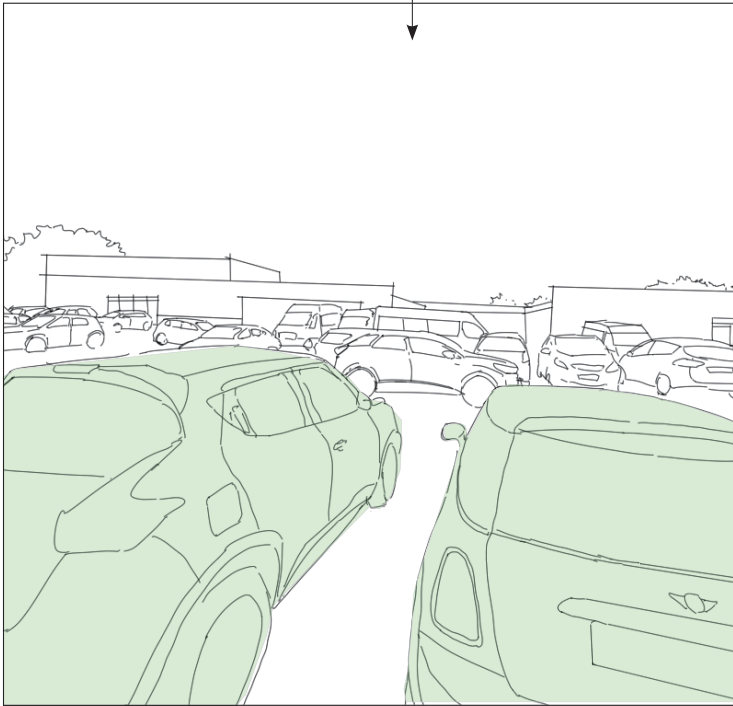


SITE SKETCH-01



The parked cars on the roadside have occupied the pedestrian path.

SITE SKETCH-01



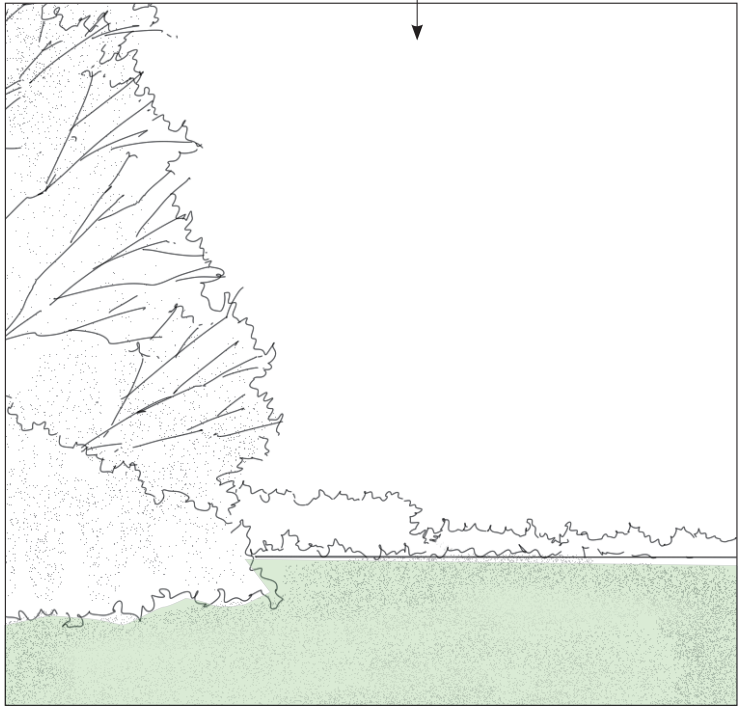
The entire campus square is occupied by parked vehicles.

SITE SKETCH-01

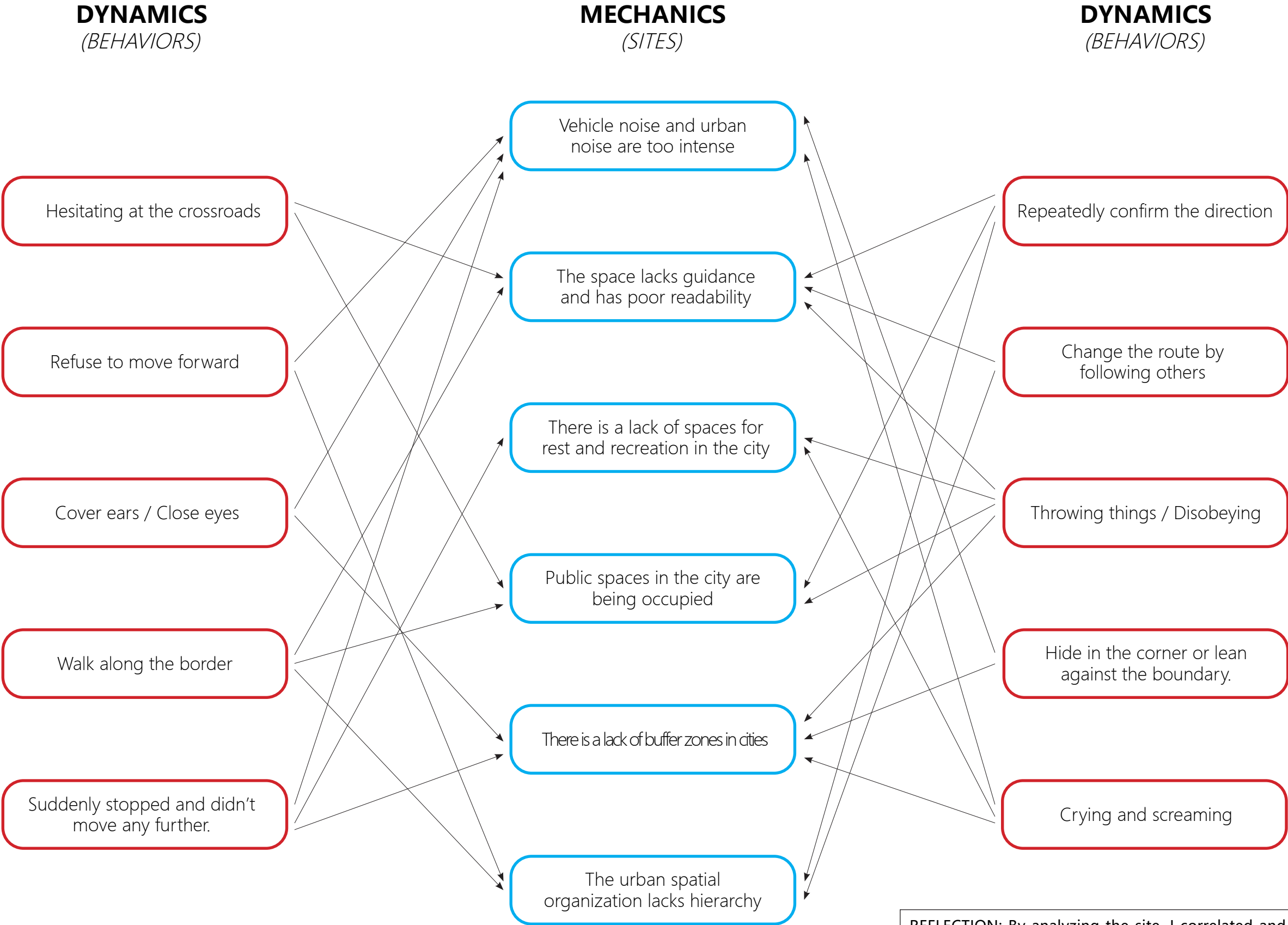


The noise near the school and the cemetery is too loud. The sidewalk is too narrow.

SITE SKETCH-01

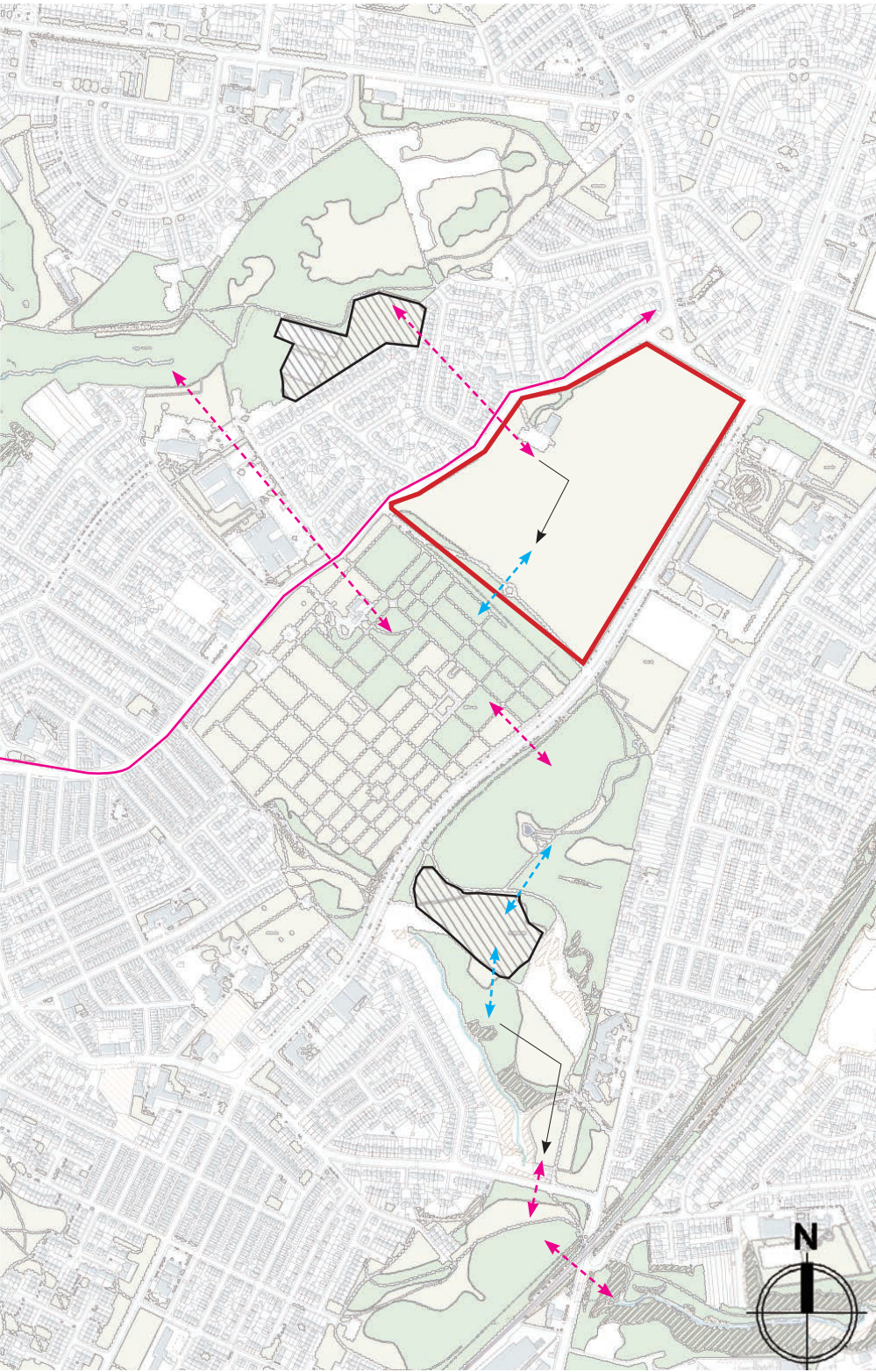


Large areas of land have become abandoned, leaving students and residents nearby with insufficient space for activities.



REFLECTION: By analyzing the site, I correlated and analyzed the Dynamics(Behaviors) and Mechanics(Site). The Dynamics do not occur independently but are interrelated with specific spatial conditions: the same Mechanic can trigger multiple behaviors, and the same Behavior can also be triggered by multiple Mechanics simultaneously. The incompleteness and imperfection of the Mechanics will affect the daily travel experience of SEN students. The design can achieve the improvement of the behaviors and emotions that SEN students may have in the site by enhancing the site Mechanics.

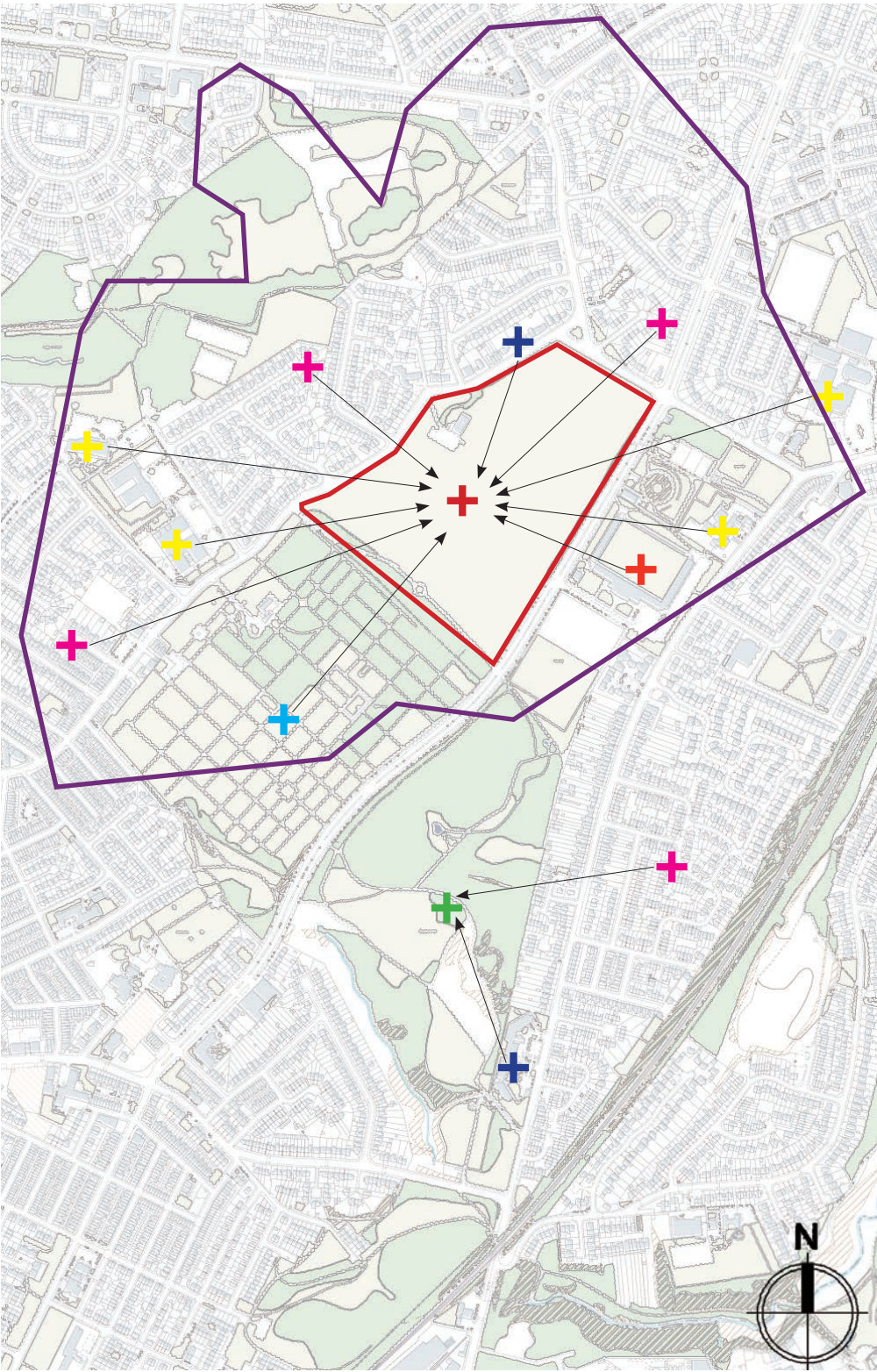
ECOLOGICAL



- LEGENDS**
- Proposal Woodland
 - Site
 - Proposal Green ways
 - Proposal Hedgerows
 - Proposal Woodland Corridors

In the designed area, the green spaces of Moston will be connected in various ways to enhance the species diversity of the original site. Moreover, by greening the urban roads, the noise generated by urban trafpc will be reduced, alleviating the sensory overload of SEN students.

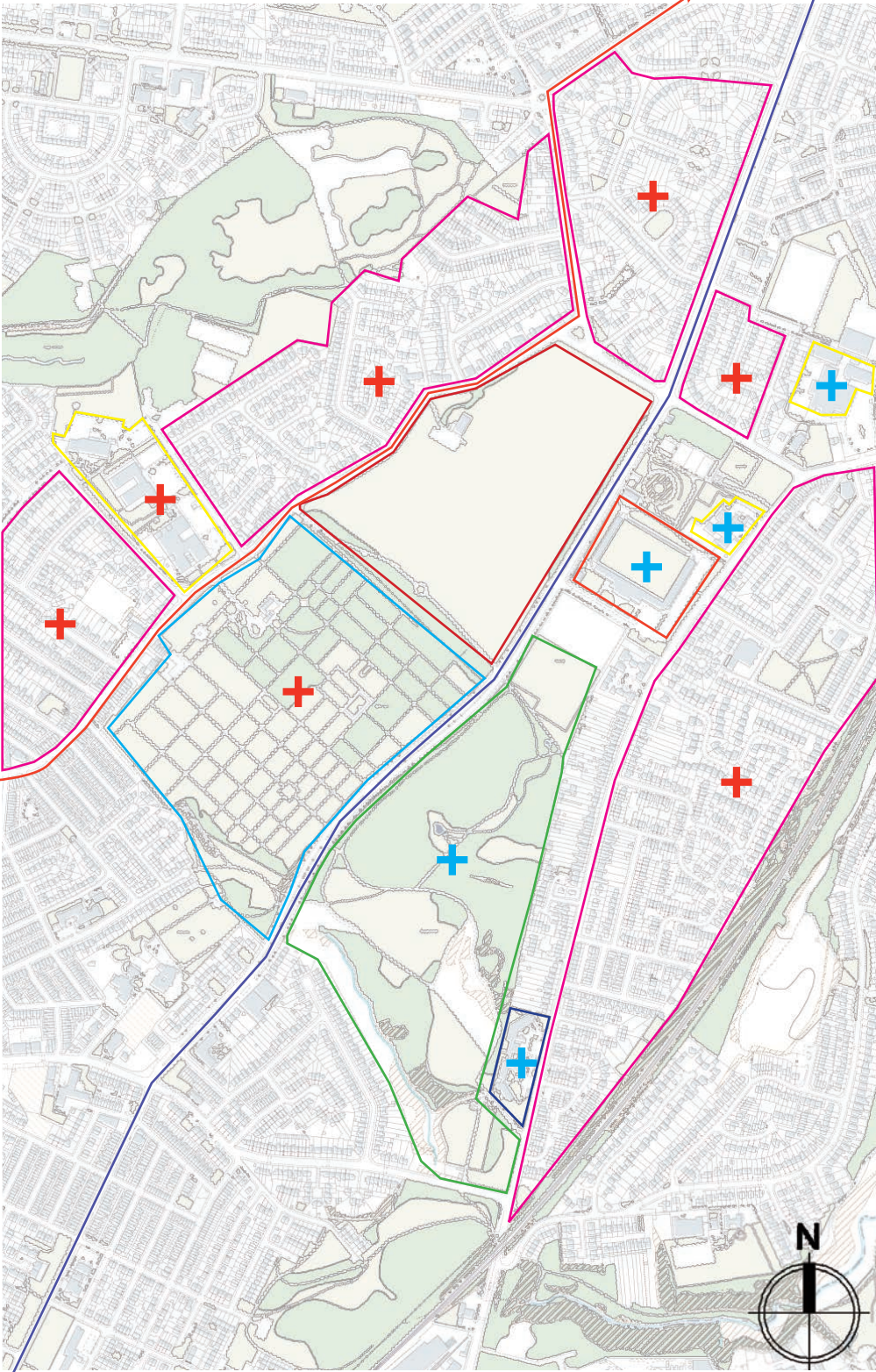
INCLUSIVENESS



- LEGENDS**
- 10-minute walking isochrone
 - Site
 - School
 - Park
 - Cemetery
 - Stadium
 - Residential Area
 - Nursing Home

The designed area will become a relaxing and recreational space within the Moston Center. It will not only serve the Sen students, but also serve as a communication and activity center for different groups of people at Moston, making the entire Moston more inclusive.

EXPERIENCE

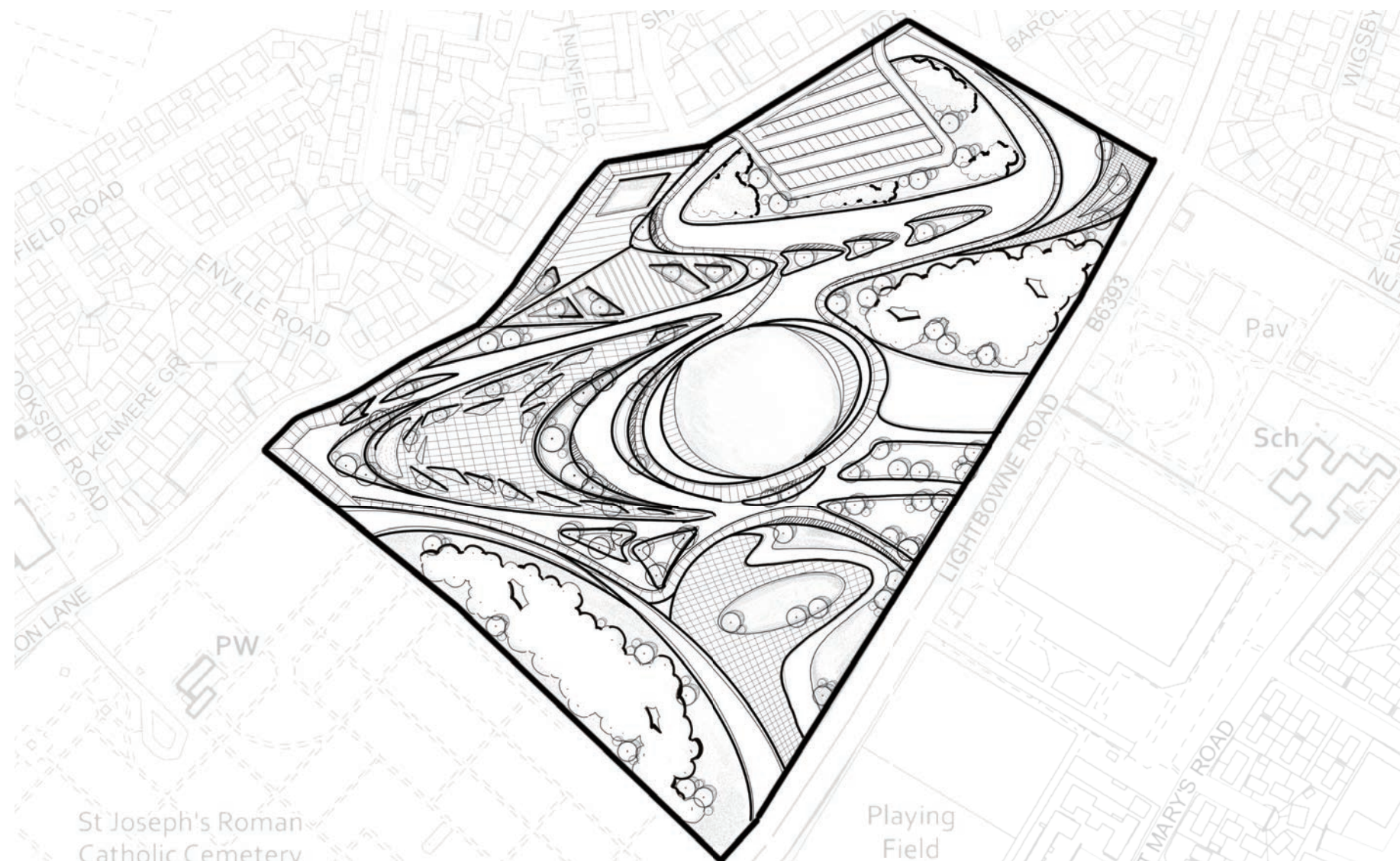


- LEGENDS**
- Residential Area
 - Nursing Home
 - Cemetery
 - Park
 - School
 - Stadium
 - Lively Route
 - Quiet Route

Around the site, we can see that the nature of the land use on the left and right sides of the site is slightly different, divided into dynamic and static areas. According to the different needs of users, different functional zones will be set up in the site to meet the usage requirements of different users.

NEXT QUESTION

After we have identified the shortcomings of the site, how can we use the design to address the problems that SEN students may encounter in the site?

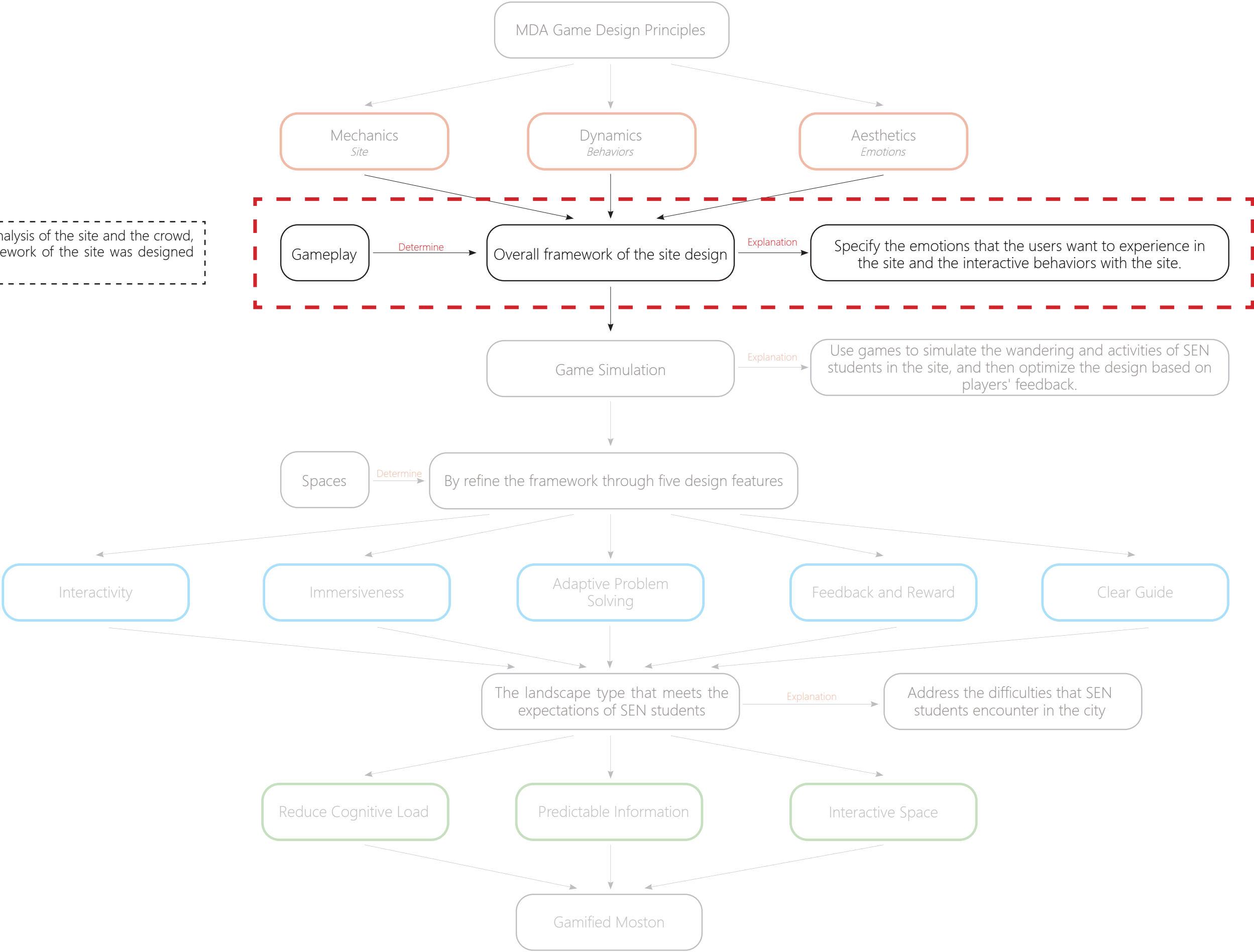


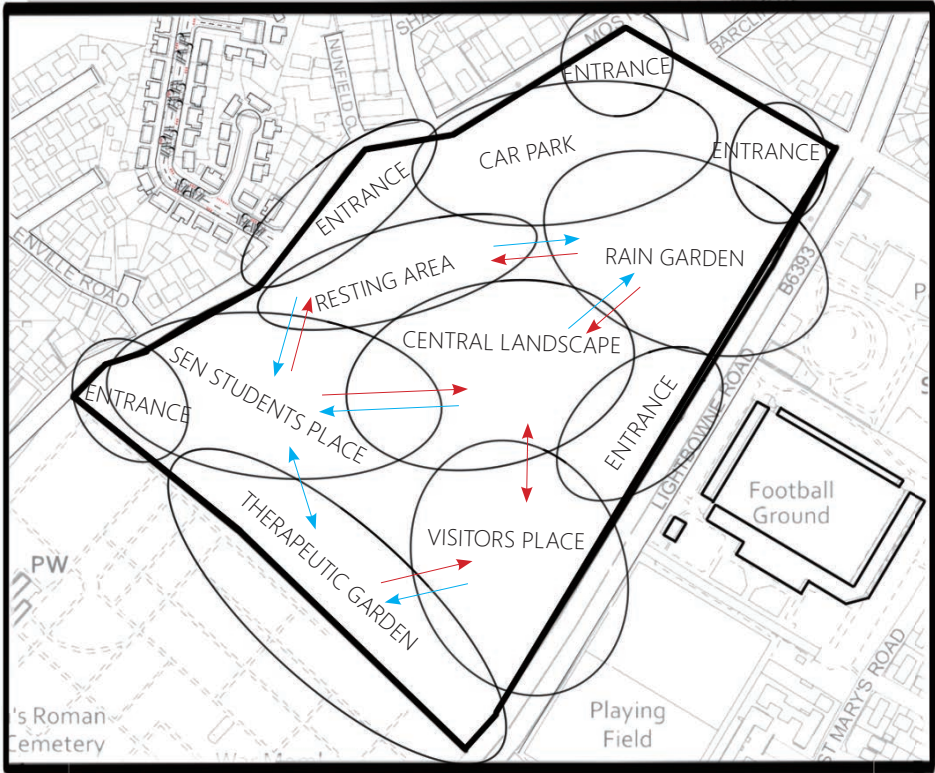
ON-PAPER DESIGN

Based on the analysis of MDA, the design initiates the paper-based design for Moston through the Mechanics of the site. By improving the Mechanics, the aim is to change the Dynamics(Behaviors) of SEN students at Moston, thereby optimizing the Aesthetics(Emotions) they experience in the city and achieving an improvement in their experience in urban spaces.

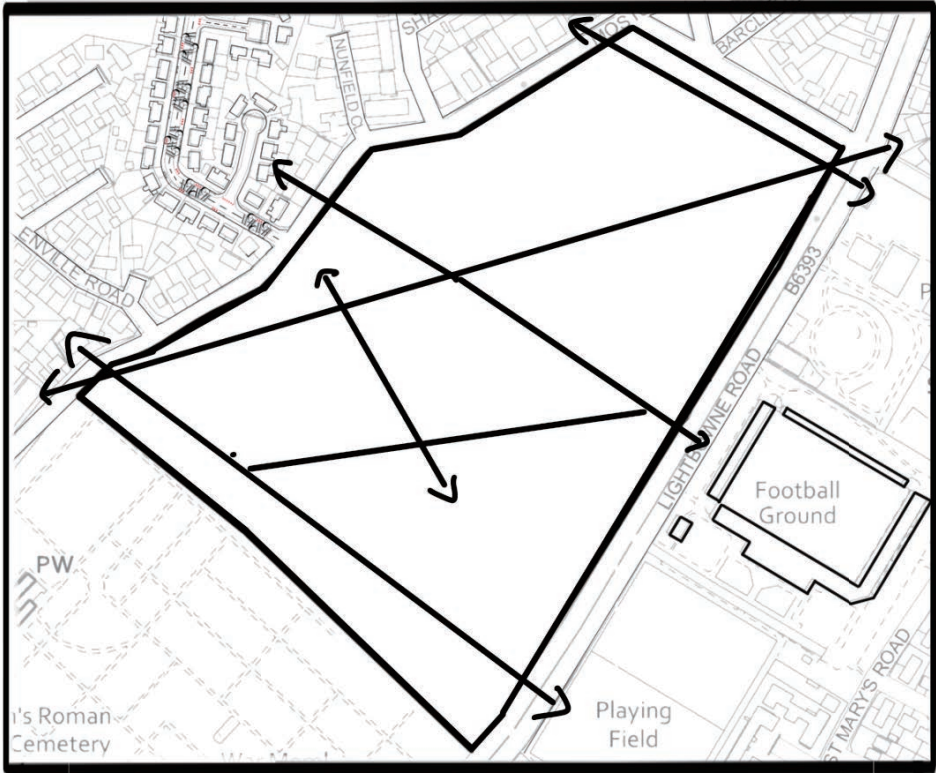
DESIGN METHOD

Based on the previous analysis of the site and the crowd, the general design framework of the site was designed and planned.

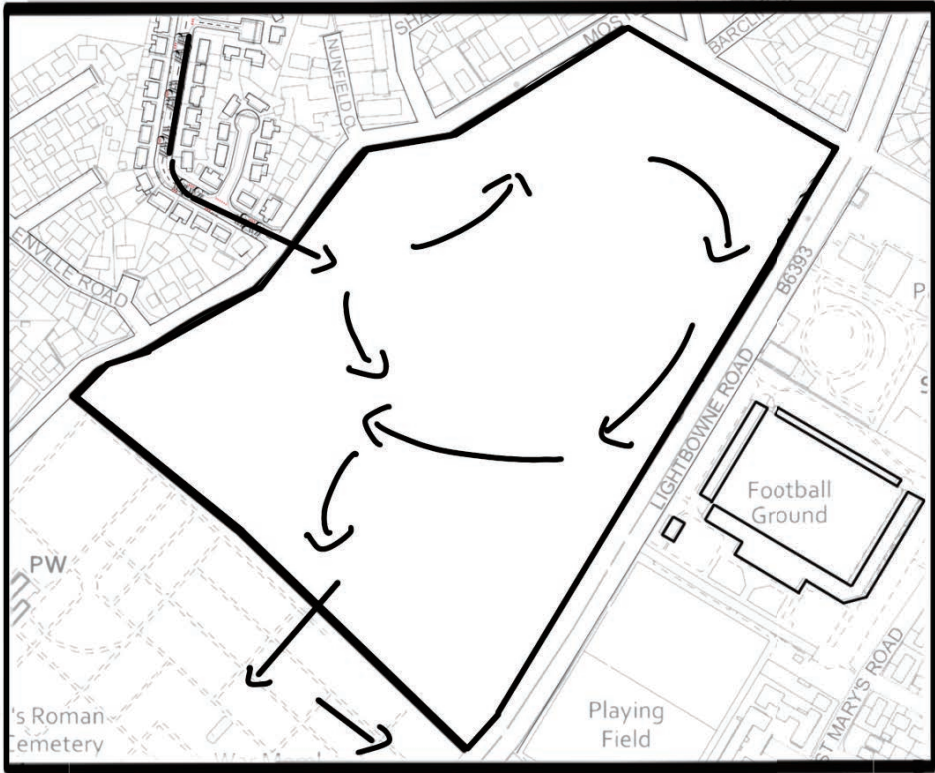




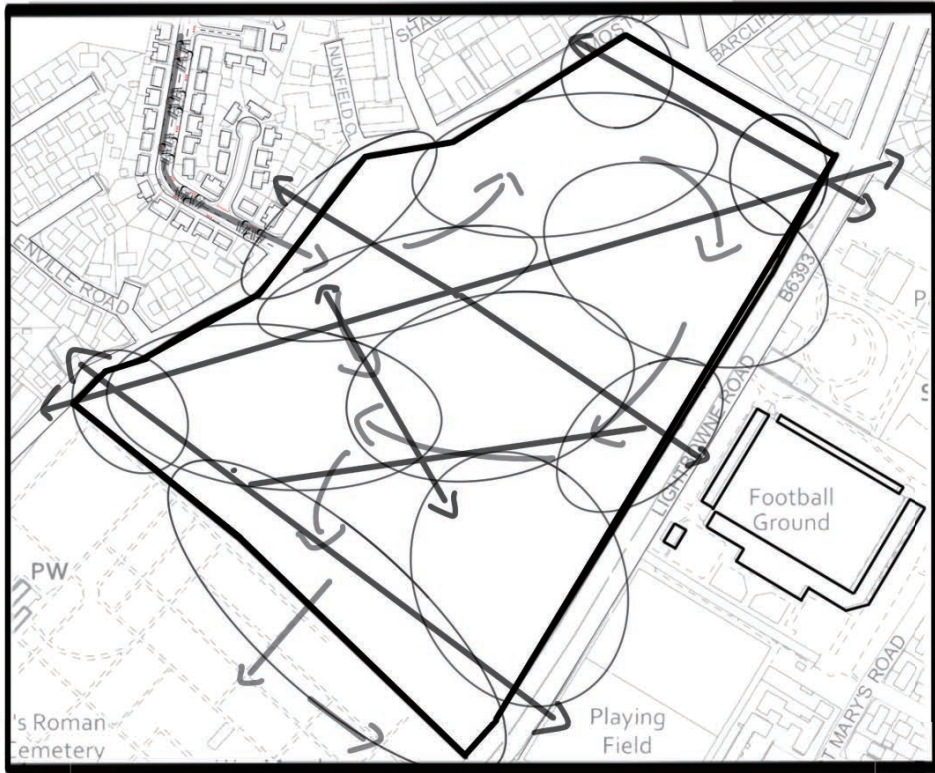
1. Divide different blocks by different functions



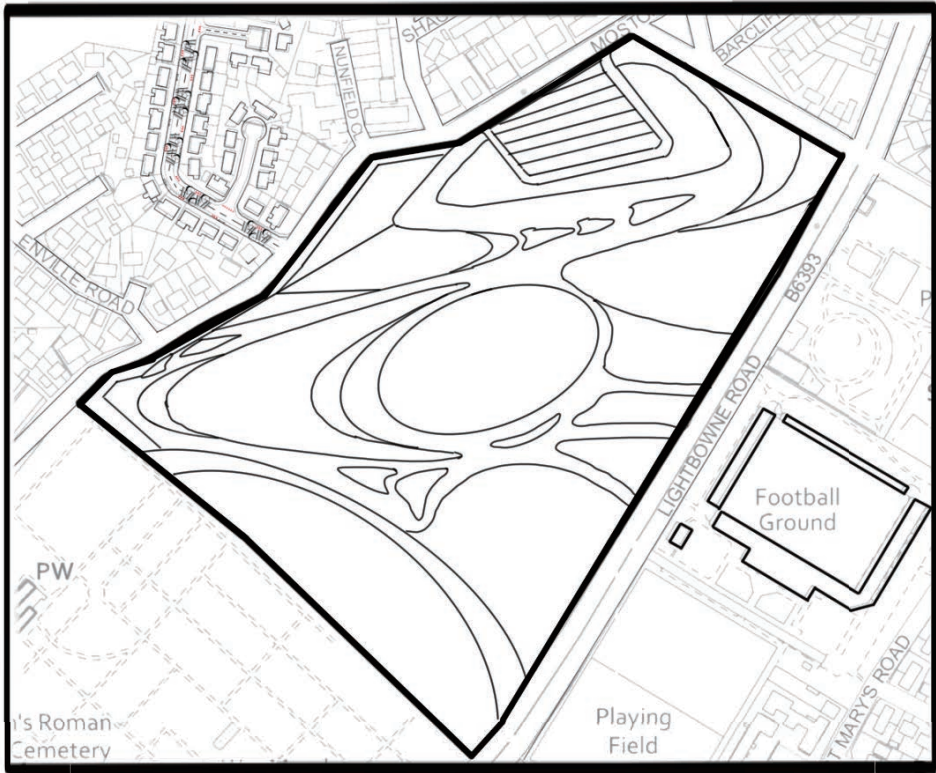
2. The propose human activity flow in the site landscape design



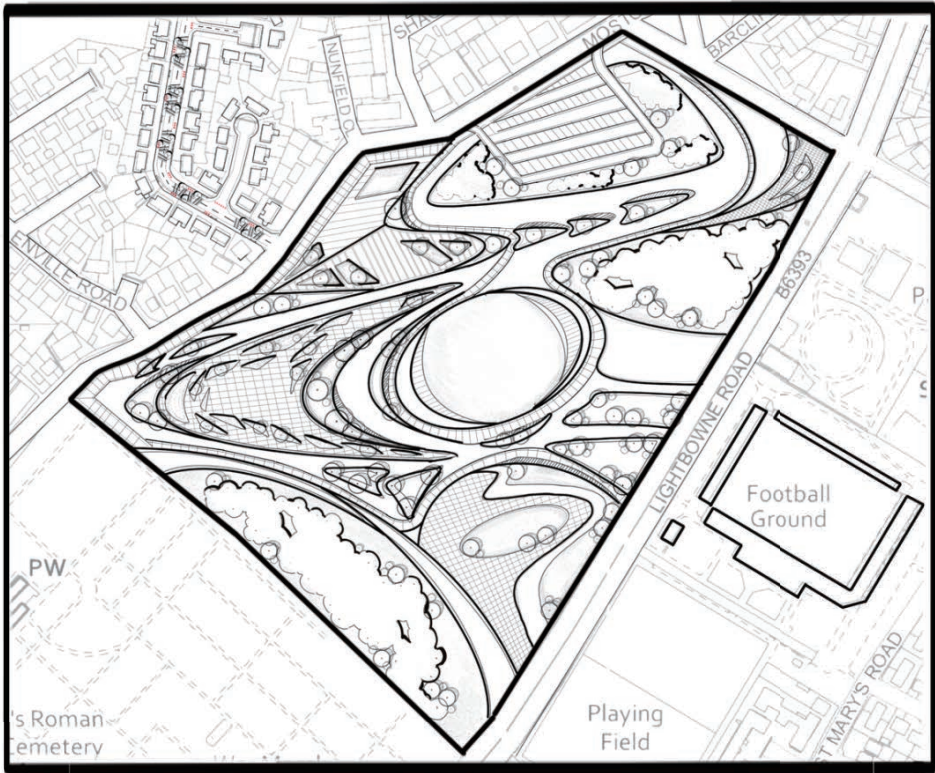
3. The propose ecological connection in the site landscape design



4. Overlay the layers



5. Design the spaces of the site according to the overlay layer



6. Refine Spaces and roads

HAND SKETCH MASTER PLAN

The available parking spaces in the parking lot are too few, and open-air parking lots tend to generate more noise. If tourists choose to park here, it will also conflict with the movement patterns of SEN students.

The central square is too open, lacking any shelter, and appears overly isolated from the surrounding environment. In the subsequent optimization process, we need to consider how to enhance the connection between the central square and the surrounding area.

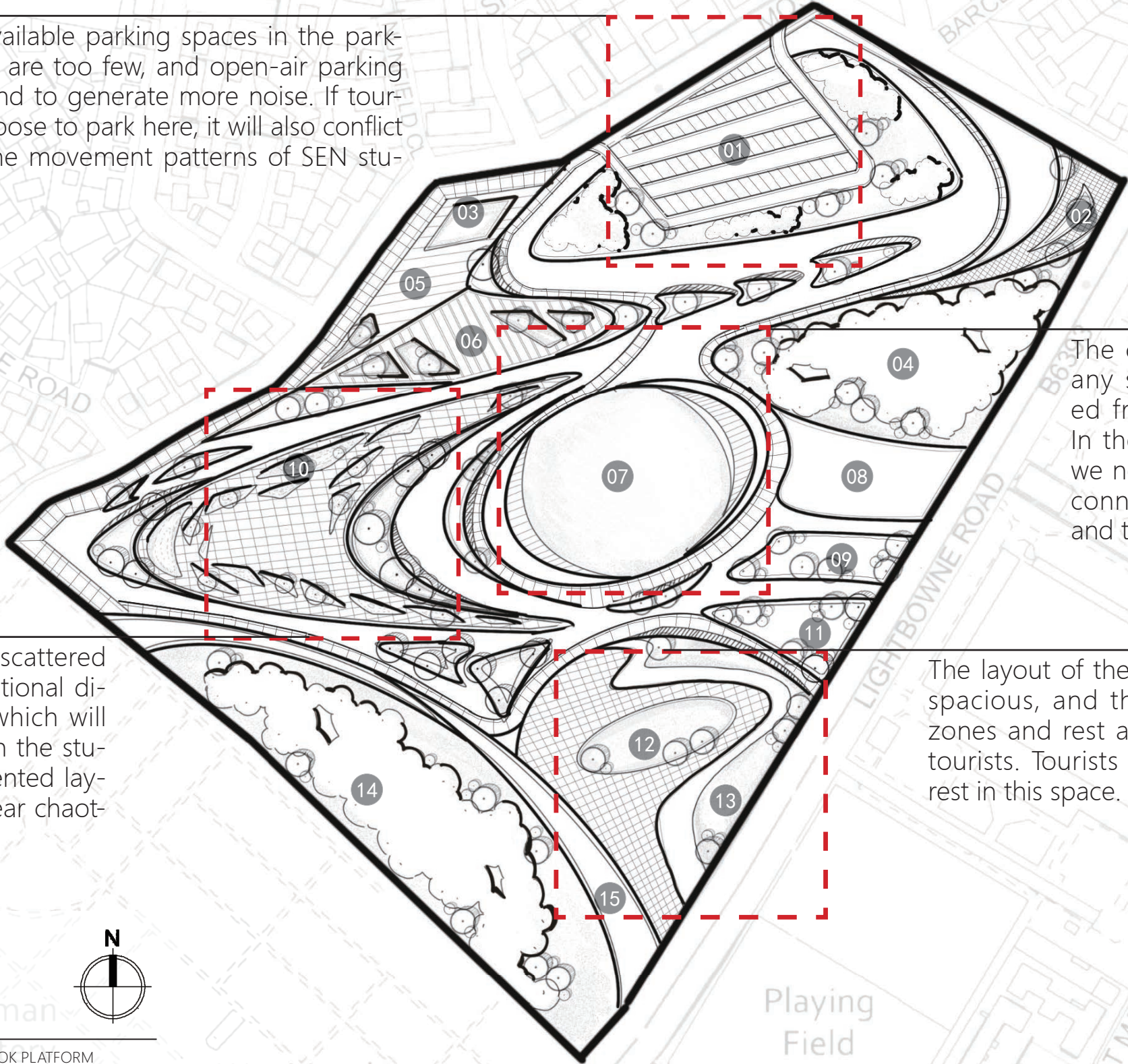
The activity venues of SEN are scattered and lack planning, and the functional divisions are not clear enough, which will increase the cognitive burden on the students of SEN. The overly fragmented layout makes the entire space appear chaotic.

The layout of the commercial area is too spacious, and there is a lack of buffer zones and rest areas near the street for tourists. Tourists may refuse to stay and rest in this space.



MASTER PLAN
LEGEND

01_CAR PARK	06_RESTING SPACE	11_OVERLOOK PLATFORM
02_ENTRANCE	07_APARTMENT TYPE 1	12_VISITORS RESTING SPACE
03_BUS STATION	08_RAIN GARDEN	13_COFFEE SHOP
04_RAIN GARDEN	09_EAST ENTRANCE	14_THERAPEUTIC GARDEN
05_MAIN ENTRANCE	10_SEN STUDENT SPACE	15_OVERLOOK BRIDGE



HAND SKETCH MASTER PLAN



MASTER PLAN

LEGEND

- | | | |
|------------------|----------------------|---------------------------|
| 01_CAR PARK | 06_RESTING SPACE | 11_OVERLOOK PLATFORM |
| 02_ENTRANCE | 07_APARTMENT TYPE 1 | 12_VISITORS RESTING SPACE |
| 03_BUS STATION | 08_RAIN GARDEN | 13_COFFEE SHOP |
| 04_RAIN GARDEN | 09_SEN STUDENT SPACE | 14_THERAPEUTIC GARDEN |
| 05_MAIN ENTRANCE | 10_SEN STUDENT SPACE | 15_OVERLOOK BRIDGE |

NEXT QUESTION

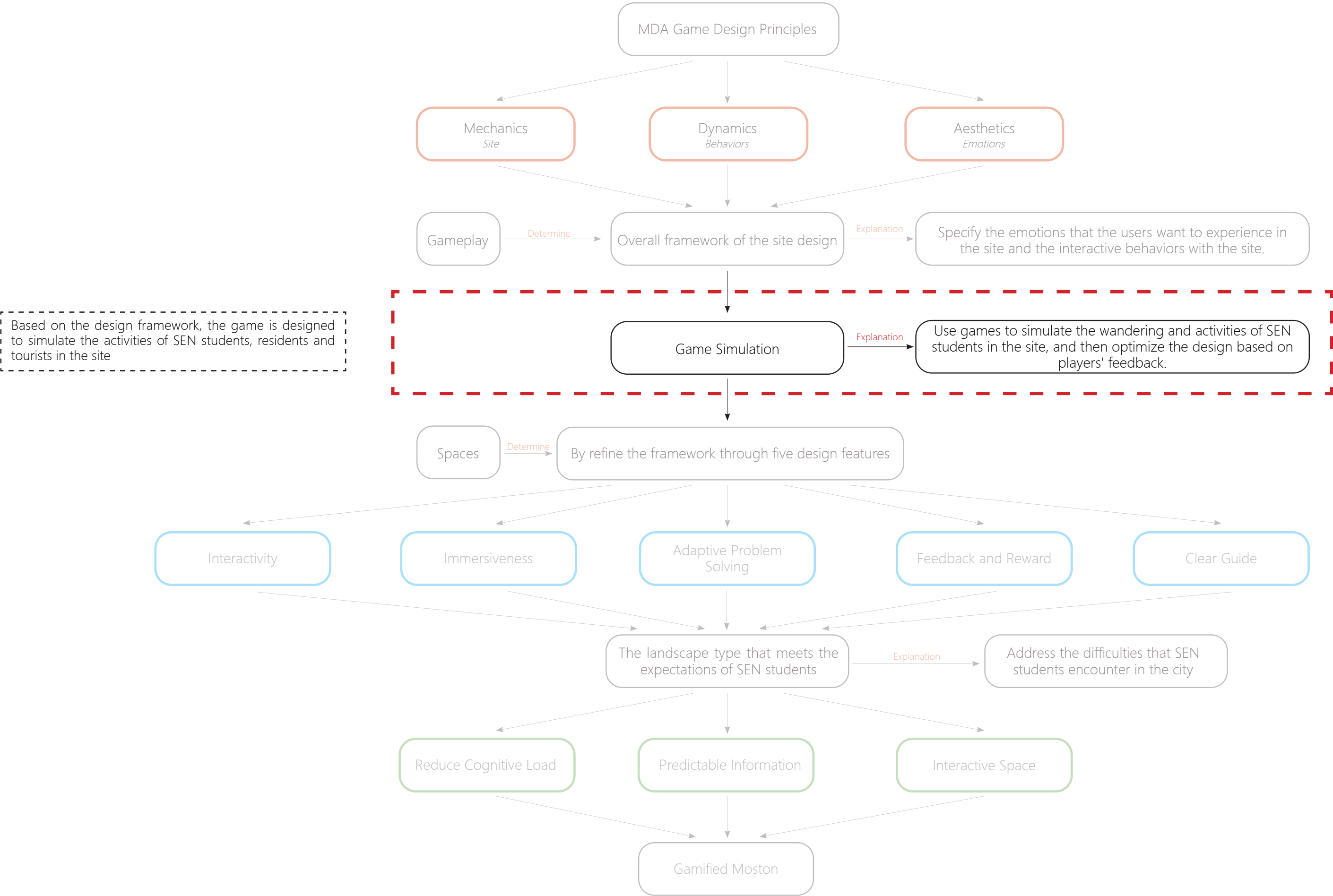
How can we identify the shortcomings of the paper-based design? How can we obtain the actual experiences of the SEN students in the designed space and optimize and iterate the design?



GAME SIMULATION

The design will be constructed step by step based on the framework, allowing more players to simulate the roles of SEN students, residents and tourists in the game. This enables them to obtain the actual feedback in the site, and based on this feedback, they will further optimize the design of the site.

DESIGN METHOD



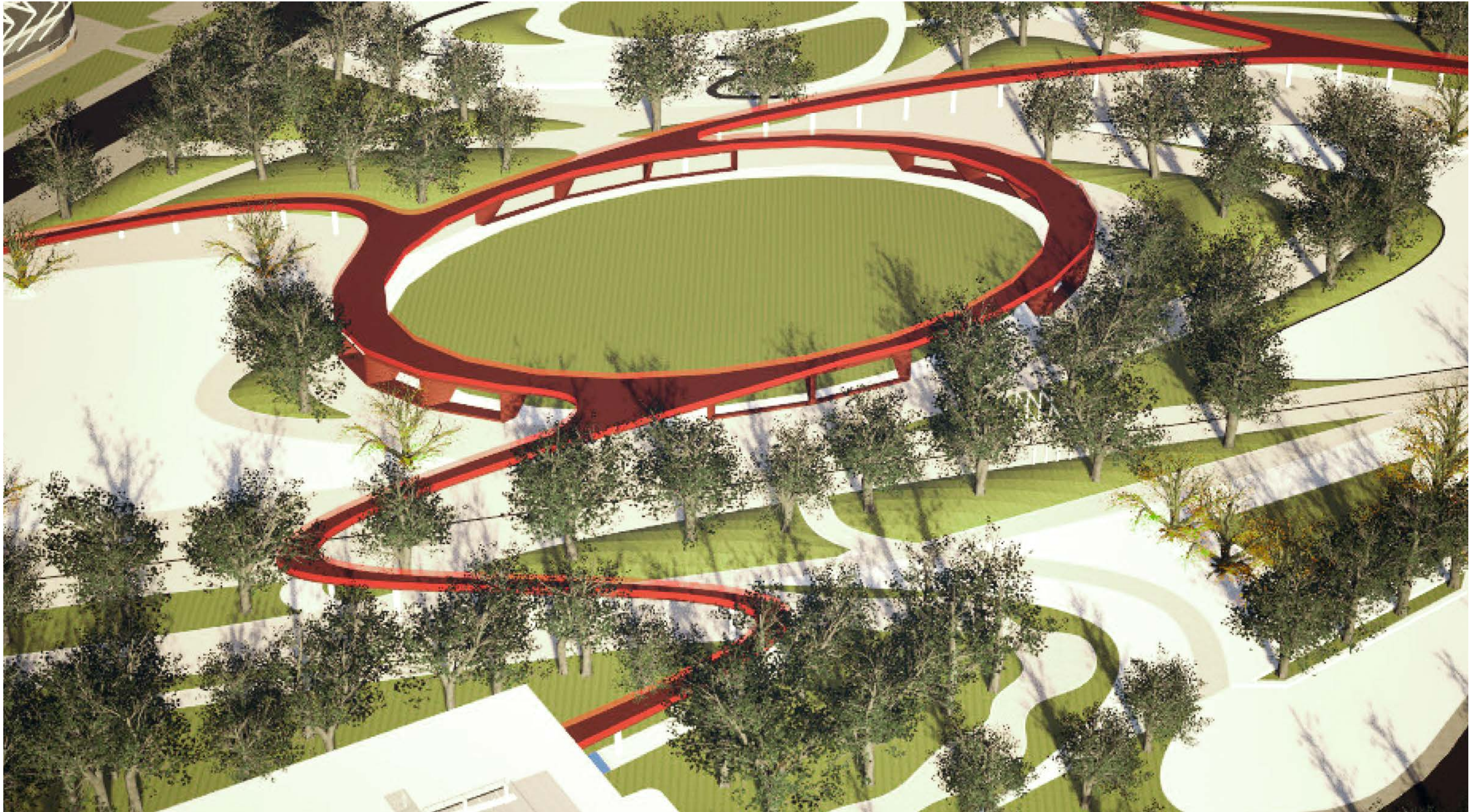
URBAN RECOVERY: CIVIC ACTION GAME *Play the City. (2026)*

Play the City adopts a game-based approach as the core collaborative decision-making method, integrating diverse stakeholders such as policymakers, market parties, and citizens into the same participation framework. It builds an efficient communication platform through low-barrier, non-technical game scenarios; replaces traditional meeting formats with game-based collaboration, enabling participants to jointly identify problems, negotiate solutions, and reach consensus during the interaction process; and combines academic research to support the entire process, converting the game outputs into implementable urban policies and spatial design results, ultimately achieving cross-group, transparent, and participatory urban decision-making and design. *Play the City. (2026)*

In the design process, I also adopted the collaborative urban gamification simulation design methodology. Based on the multi-party governance game mechanism of "Play the City", combined with the MDA game design theory, I constructed an inclusive landscape design path for students with special needs (SEN). By setting up a role-playing site simulation game, multiple users could assume the roles of SEN students, residents, tourists, etc. During the immersive game process, they could identify immediate needs and provide feedback on real experiences. Based on the behavioral trajectories, emotional feedback and path choices in the game, spatial obstacles such as noise, lighting, guidance, and accessibility were identified. Then, through iterative game optimization, the consensus reached by the multiple parties through collaboration was transformed into specific design strategies, ultimately achieving landscape space co-creation and optimization driven by gamification collaboration, based on real experiences, and with the goal of inclusivity and accessibility.



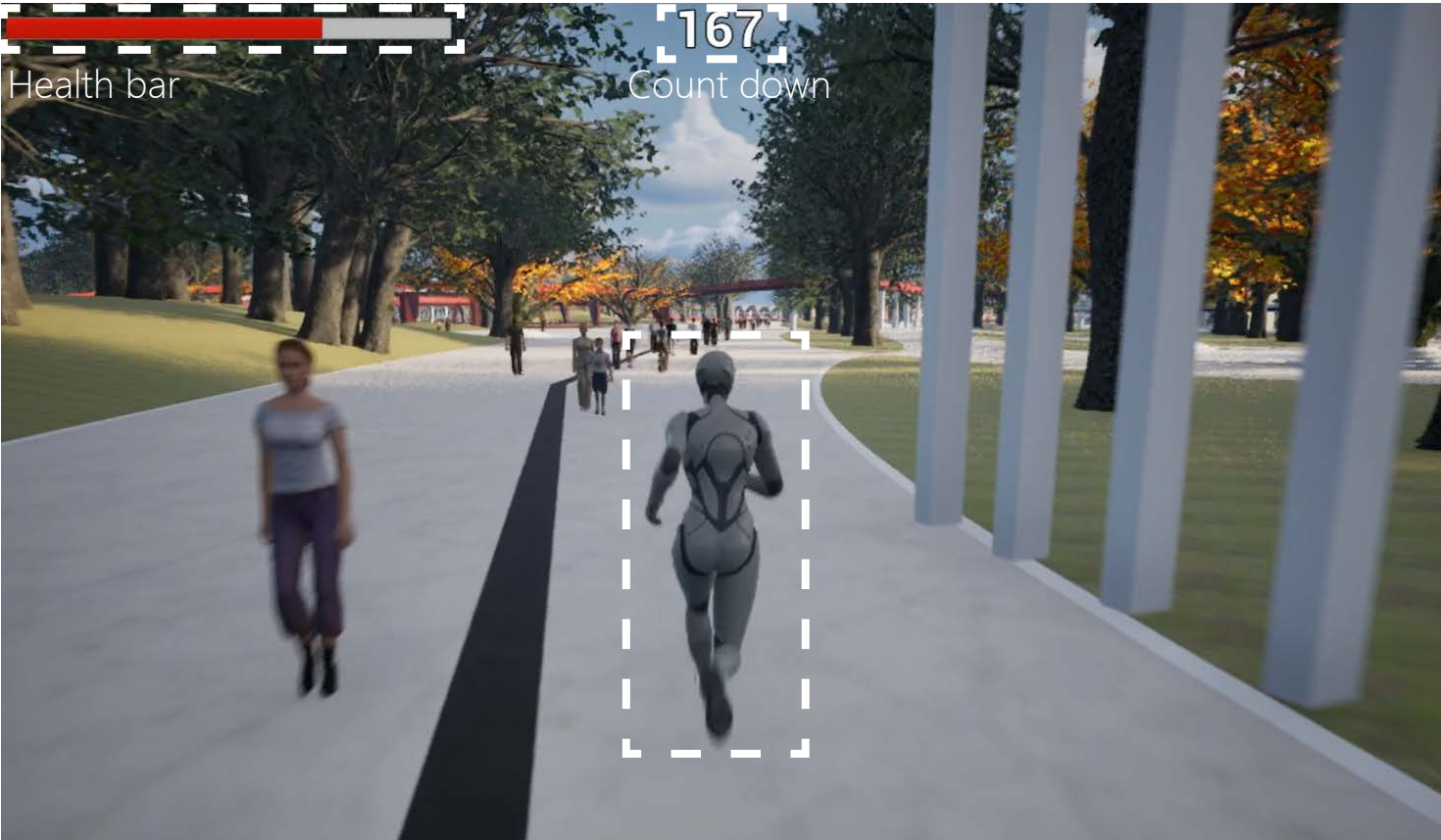
Play the City. (2026)



Download Link: https://drive.google.com/ple/d/1P1vhbJ_ofNI7D4UrxHVKBHkEbl38yZrj/view?usp=sharing

In the video, I attempted to use sound to simulate the core issue of autism in the urban space of Moscow. They live in a noisy environment where the honking of car horns and the rumbling of trains constantly stimulate the nerves of people with autism, making it difficult for them to live in the city.

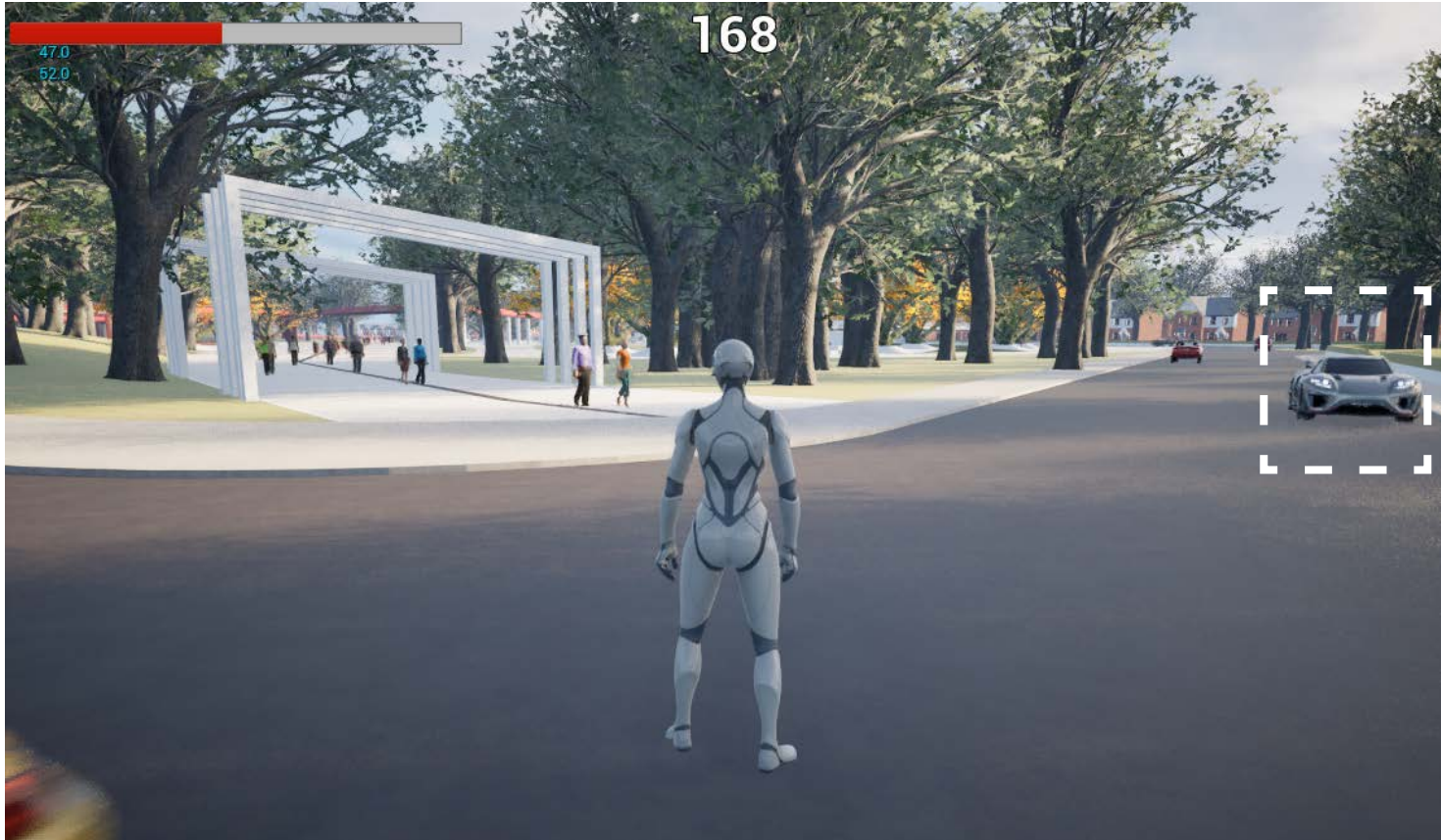
GAMEPLAY By using gameplay to simulate the sensory overload characteristic of SEN students in the environment



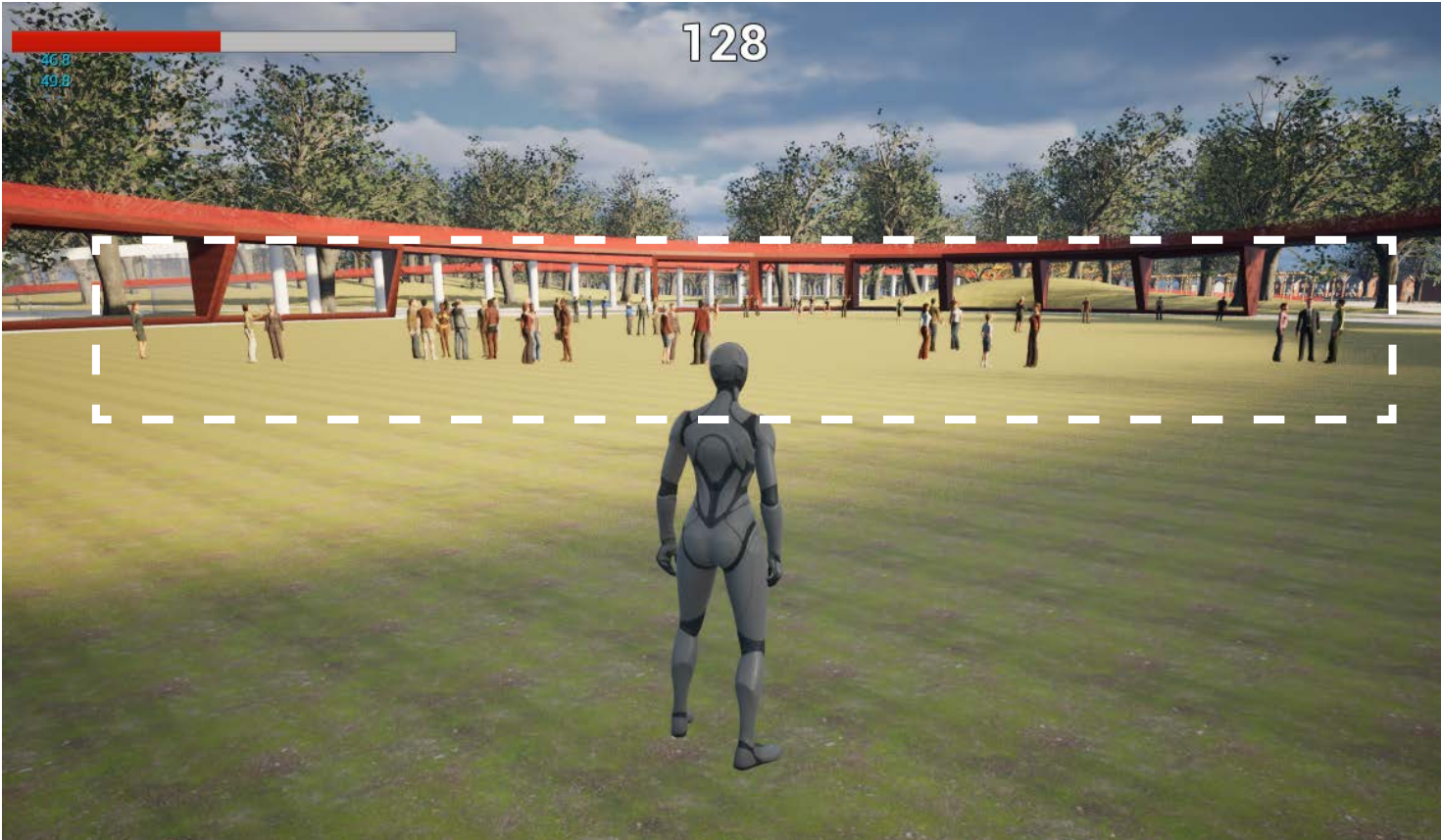
TIME-LIMITED TRIP TO SCHOOL: When the health bar runs out or the countdown ends and the player fails to reach the school, the game is considered a failure.



AFRAID OF THE DAZZLING SUNLIGHT: When the player stands in the sunlight, the screen will gradually become illuminated and the health points will be deducted.

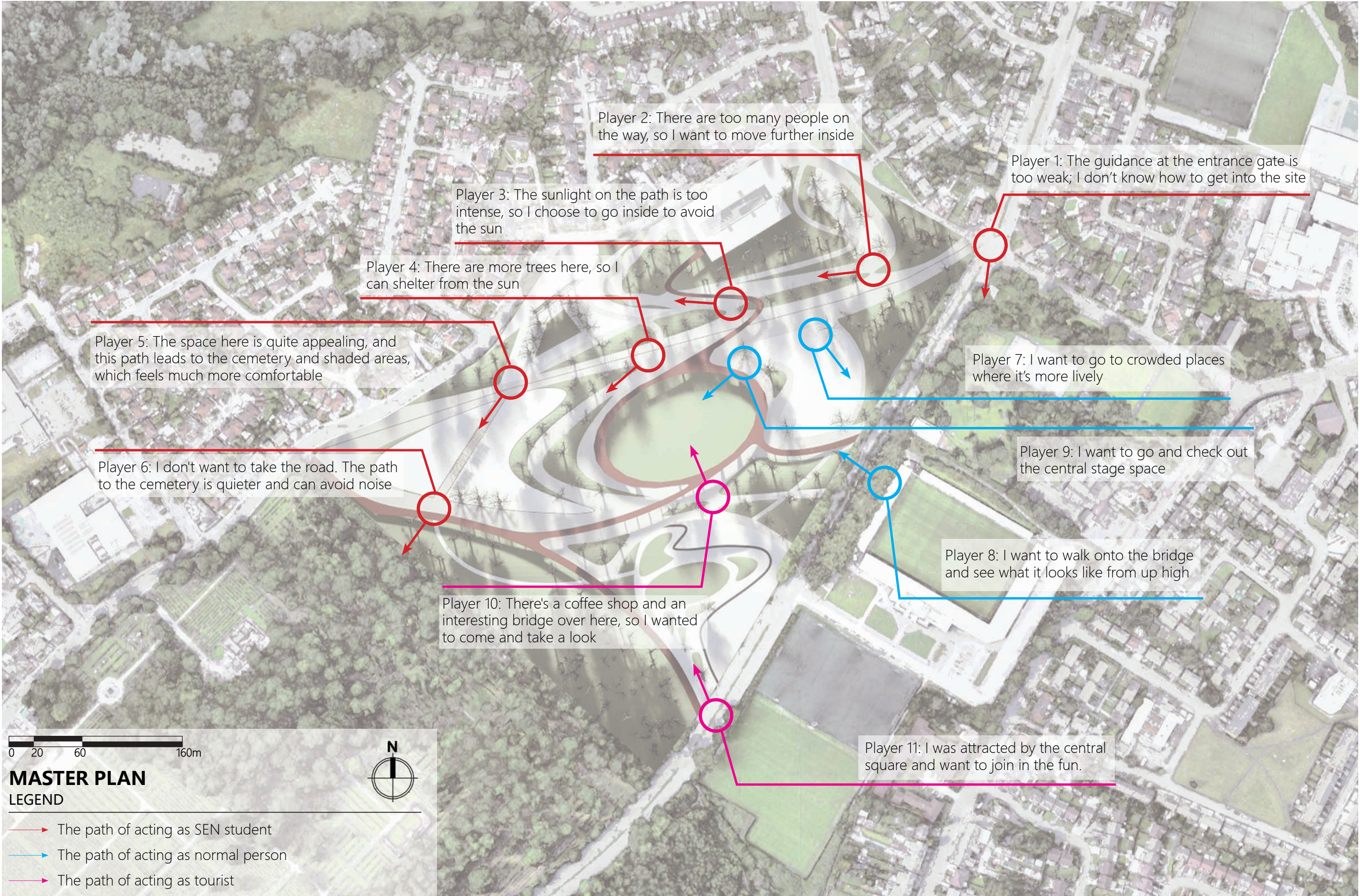


AFRAID OF NOISY SOUNDS: When the player is near the road, their health will be rapidly depleted.

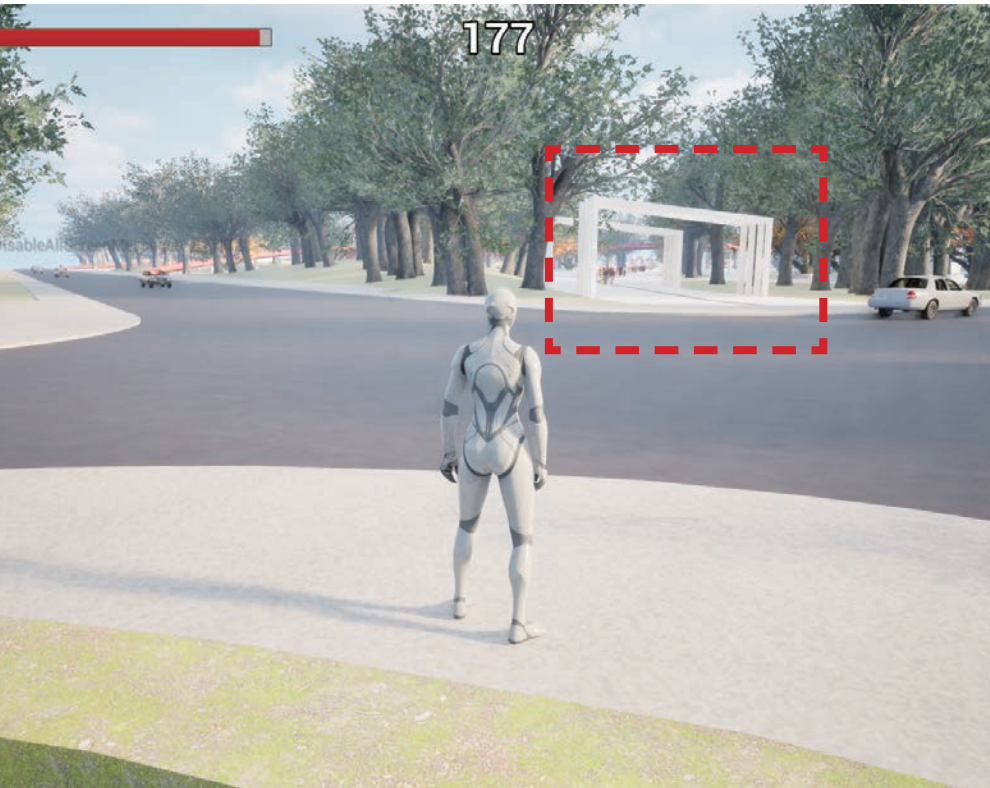


AFRAID OF CROWDS: When the player approaches the crowd, their health will be rapidly depleted.



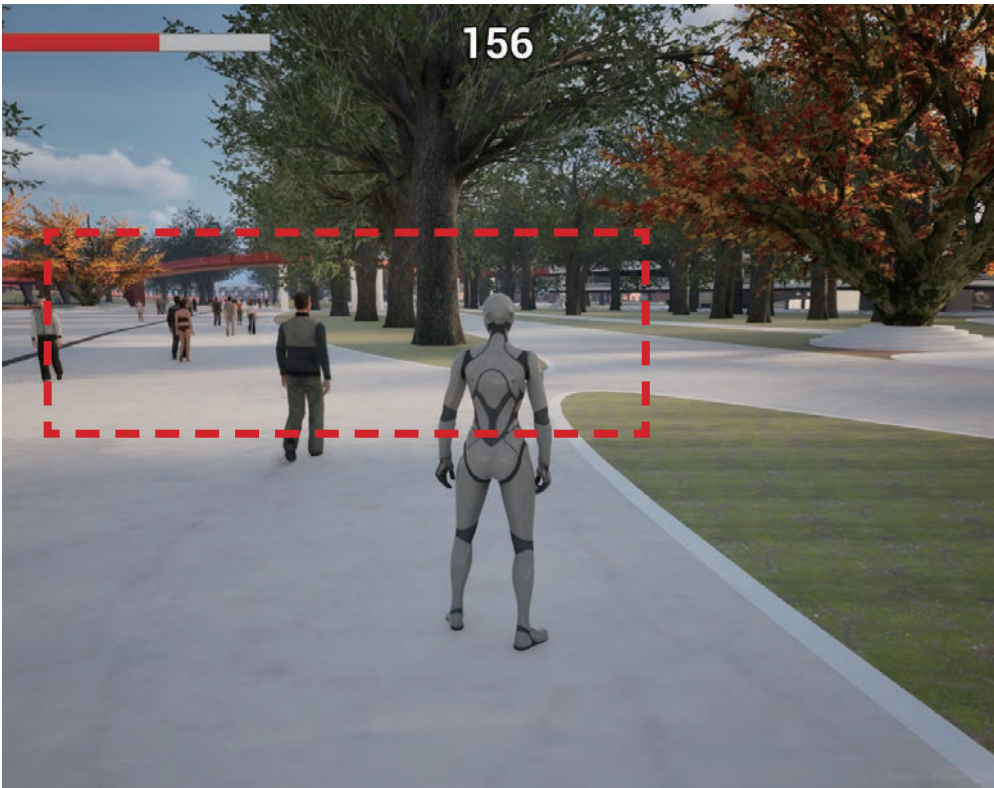


ENTRANCE



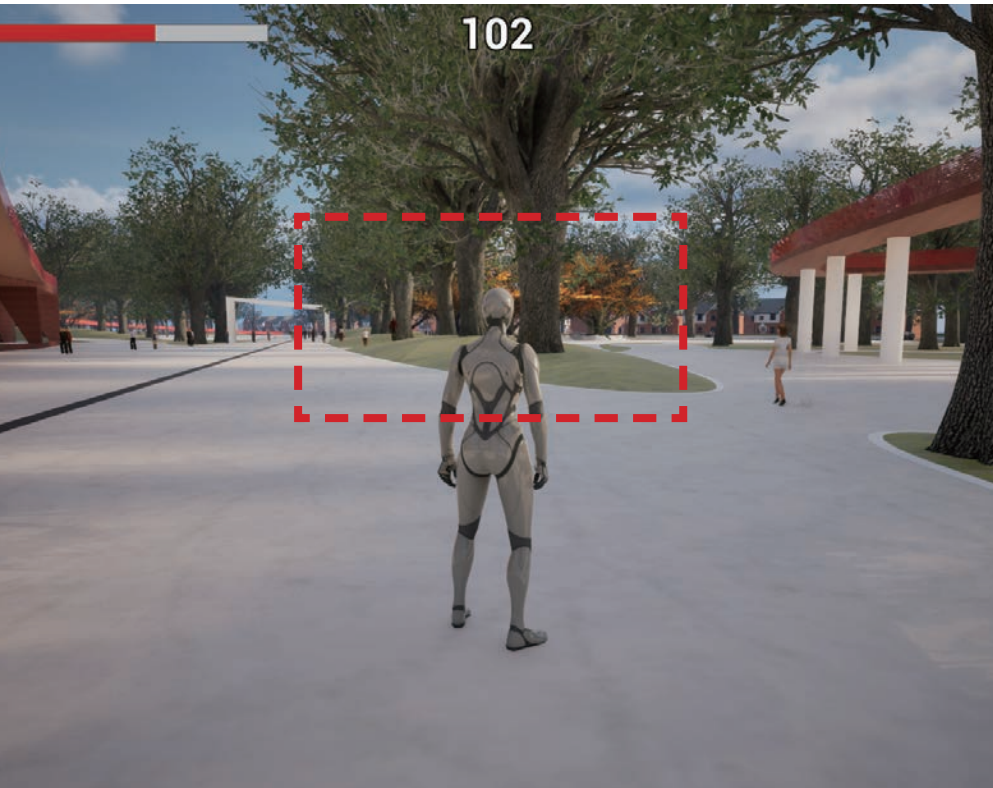
Strengthen the guidance of the site to make it easier for SEN students to know the direction of the site and the school.

MAIN ROAD 01



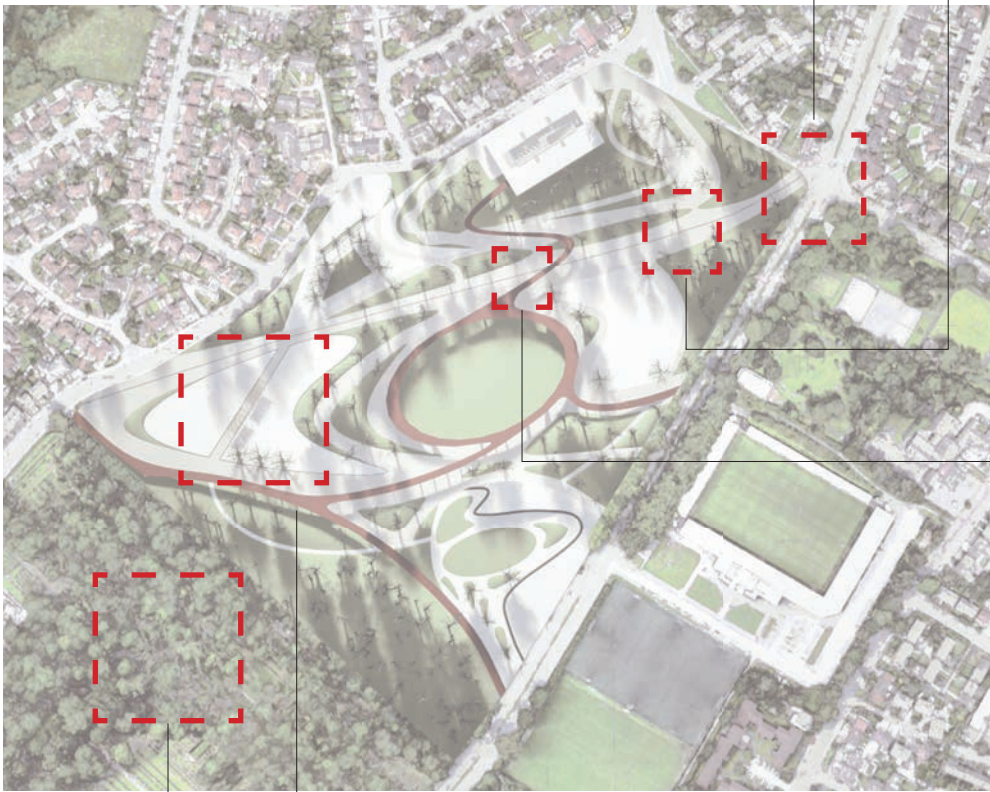
The sunlight on the main road is too intense, which can make the planting of plants on the main road more diverse and add some pavilions for shade.

MAIN ROAD 02

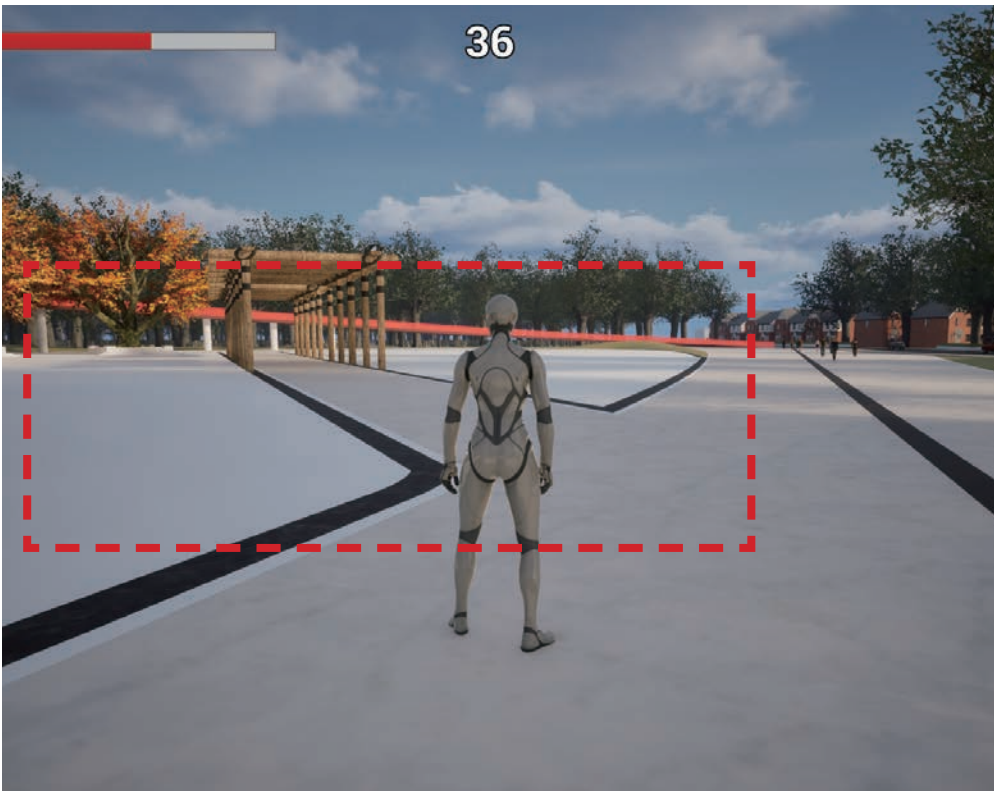


Designing rest points on the road can provide rest areas for SEN students and alleviate the harm caused by sensory exposure.

MASTER PLAN

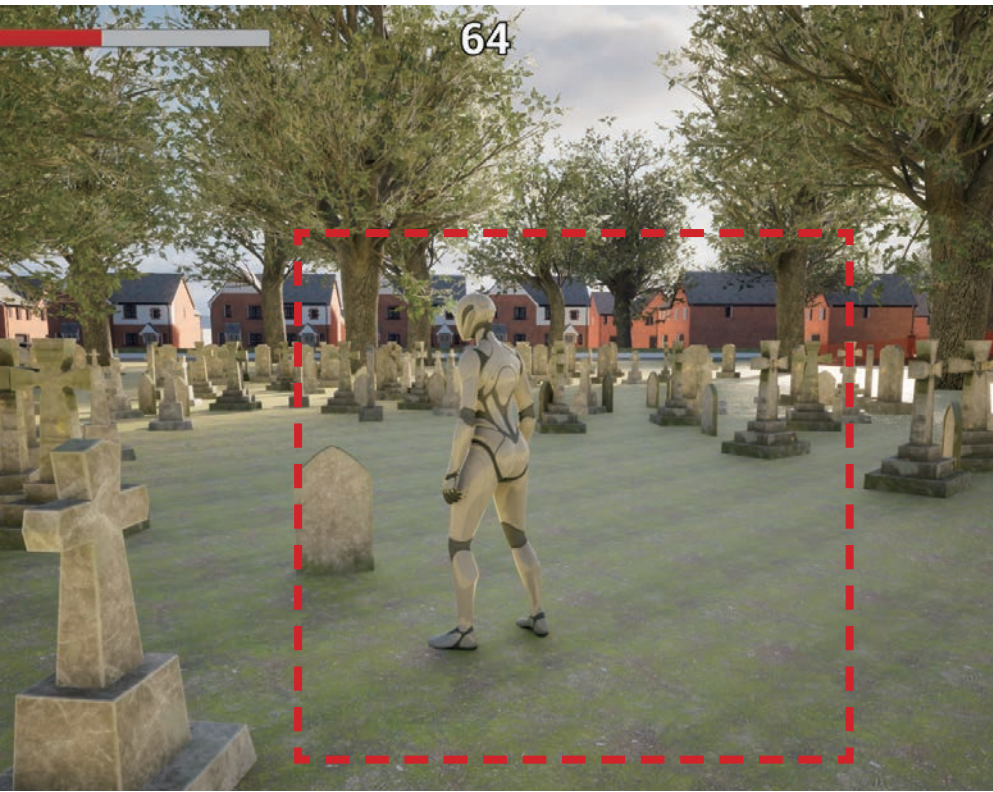


SEN STUDENT PLACE



As the area that is most frequently visited by SEN students, it can be used as a rest area and activity space for them.

CEMETERY



Many players who play as the SEN students will go to the cemetery to avoid the noise and sunlight. This can optimize their walking experience in the cemetery.



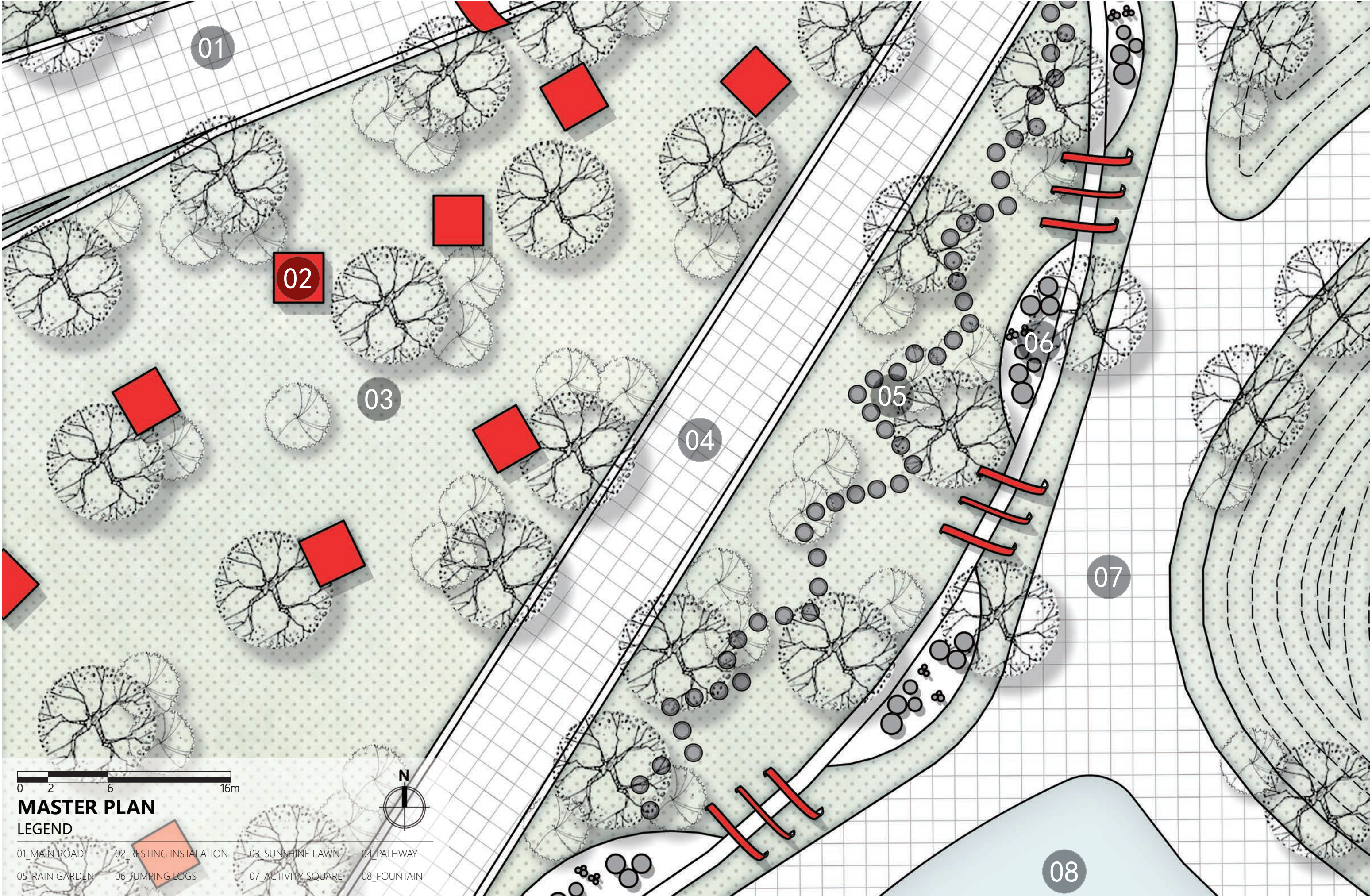
COMER CHILDREN' S HOSPITAL PLAY GARDEN *Sohu. (2021)*

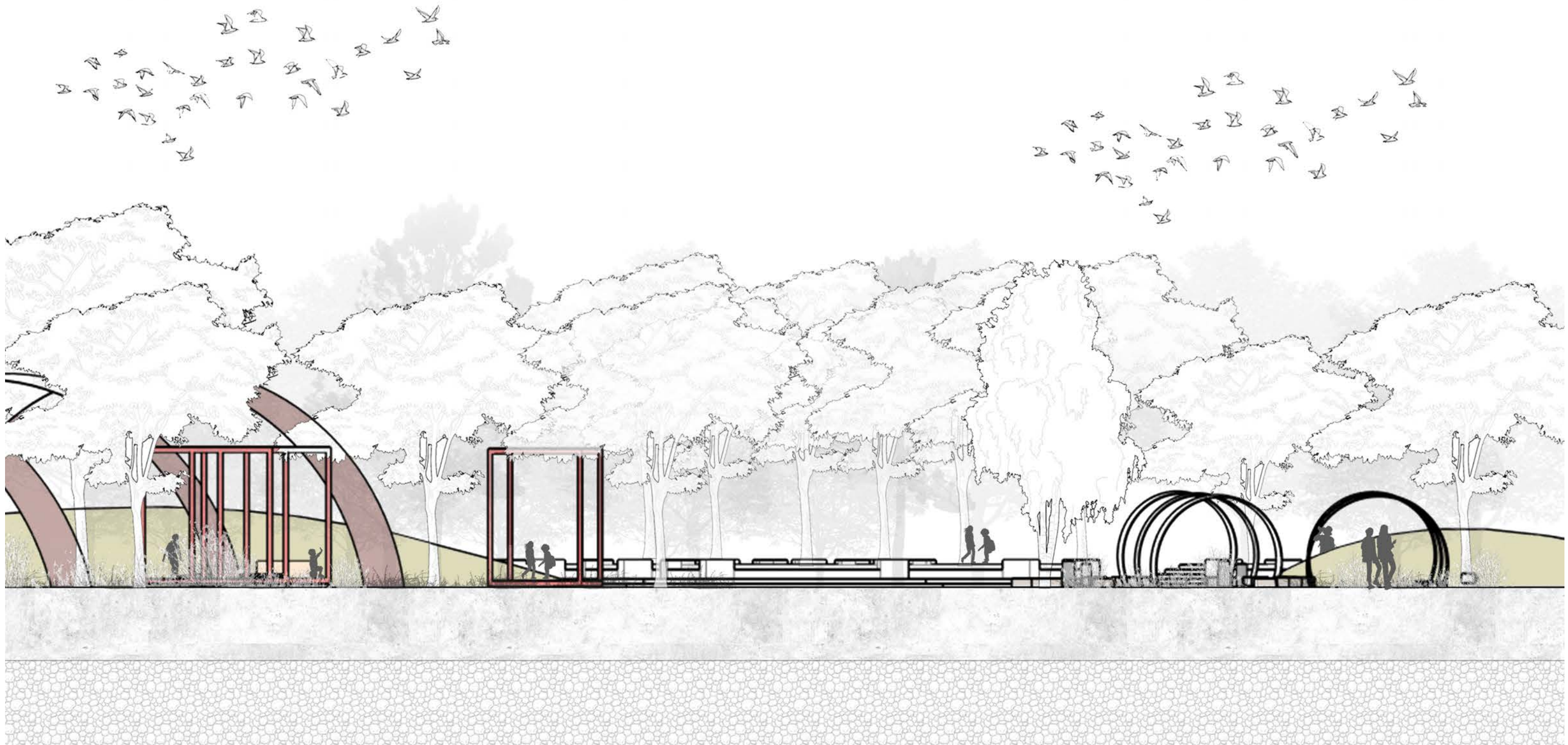
The Game Healing Garden of Comer Children's Hospital in Chicago is designed with the core concepts of full-age and full-ability adaptability, multi-sensory play experiences, miniature natural healing, and safe and private small spaces. It integrates barrier-free access, tactile, auditory and visual lightweight interactions, natural plants and rest nodes, providing a low-stimulation, controllable and self-participatory healing environment for children with limited mobility and sensory sensitivities and their families, achieving a combination of the joy of play and rehabilitation assistance. *Sohu. (2021)*



Sohu. (2021)

I applied the healing landscape and gamification design concepts from this case to the design of key areas in this project. By setting up continuous barrier-free flow lines, low-load sensory interaction nodes, natural shading and noise-reducing plant designs, I reduced the sensory overload and anxiety of SEN students. At the same time, by drawing on its lightweight game mechanism, I added touchable, listenable designs and rest-able landscape structures. Under the premise of ensuring safety and accessibility, I enhanced the inclusiveness, emotional soothing ability and gamification experience of the space, making the key area truly become a core place that supports SEN students' independent activities and emotional regulation.





CONCLUSION

During this semester, my work focused on Moston, with the core being the application of gamification design to urban public spaces to enhance the **inclusiveness, interactivity, and social integration** of students with special educational needs (SEN).

The project was based in the Moston area of Manchester, UK. Initially, through methods such as heart rate graphs, noise maps, first-person recordings, and sensory analysis, the project revealed the sensory overload, lack of direction, poor accessibility, crowded spaces, and environmental stress that were causing exclusion for SEN students in the local streets, school routes, and open spaces. **Based on the MDA game theory, a site research method was established to uncover the inconveniences faced by SEN students in the site, as well as their causes and solutions.**

Through **role-playing games**, users were allowed to act as SEN students, residents, and tourists, and they provided feedback on paths, anxiety, obstruction, guidance, etc., in the game, which was used to optimize the space. Finally, based on the feedback from the game simulation, the design was optimized to transform the daily urban space into one that can simultaneously promote communication among diverse groups of people and achieve a more fair, inclusive, and supportive urban space for the healthy travel and social participation of SEN students.

REFERENCE LIST

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